

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-32001-AK.pvdb

Date of Analysis : Jul 4, 2026 11:00am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1900.000 mm ID
 Vessel Design Length, Tangent to Tangent 1323.83 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 180 °C
 Channel Side Design Temperature 80 °C
 Shell Side Design Pressure 23.454 kgf/cm²
 Channel Side Design Pressure 10.197 kgf/cm²
 Shop Shell Side Test Pressure 31.247 kgf/cm²
 Field Shell Side Test Pressure 34.060 kgf/cm²
 Shop Channel Side Test Pressure 13.584 kgf/cm²
 Field Channel Side Test Pressure 13.677 kgf/cm²

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-516 60	...	...	K02100	
Flange	SA-350 LF2	1	...	K03011	+IT
Flange	SA-516 70	...	...	K02700	
Nozzle	SA-333 6	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-334 6	...	...	K03006	+IT
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-320 L7	...	<= 65	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS CH Cover	10.407	0.00	10.500	3.0000	N/A	No
LHS Ch Flg	10.387	0.00	10.400	3.0000	N/A	No
LHS Ch Shell	10.388	0.00	18.500	3.0000	N/A	No
Main Shell	23.568	1.05	24.000	1.5000	N/A	No
RHS Ch Shell	10.388	0.00	18.500	3.0000	N/A	No
RHS Ch Flg	10.387	0.00	10.400	3.0000	N/A	No
RHS Ch Cover	10.407	0.00	10.500	3.0000	N/A	No

Liquid Level: 210.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	12.3	12.0	128.0	120.0	116.5	...	0.85	0.85



Body Flg	27.4	15.1	143.0	115.0	113.3	...	0.85	0.85
Cylinder	227.4	200.0	18.0	18.0	11.3	...	0.85	0.85
Cylinder	1091.4	858.0	18.0	18.0	17.6	...	1.00	1.00
Cylinder	1297.4	200.0	18.0	18.0	11.3	...	0.85	0.85
Body Flg	1312.8	15.1	143.0	115.0	113.3	...	0.85	0.85
Body Flg	1323.8	11.0	143.0	110.0	108.0	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	120	35.7287	...	No Calc
20	30	115	94.3864	...	No Calc
30	Tsht	18	No Calc	...	8.31
Tsht	40	18	No Calc	...	No Calc
40	Tsht	18	13.1532	1.05461	2.47
Tsht	50	18	No Calc	1.05461	No Calc
50	60	18	No Calc	...	8.31
60	70	115	92.6592	...	No Calc
70	80	110	32.9229	...	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	215.097	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	858	2037.44	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	215.103	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2-Tube Inlet	WRC-297	0.992	Shell	Passed
N1-Shell Inlet	WRC-297	0.992	Shell	Passed
N3-Shell Liqui	WRC-297	0.992	Shell	Passed
N4-Shell Vapou	WRC-297	0.992	Shell	Passed
N5-Tube Outlet	WRC-297	0.989	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	415.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1800.000 mm
Wear Pad Width	625.000 mm
Wear Pad Thickness	20.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1000.000 mm

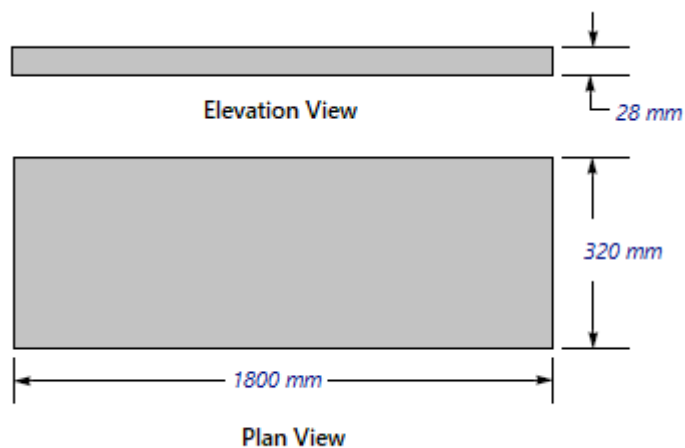
Baseplate Length	1800.000 mm
Baseplate Thickness	28.000 mm
Baseplate Width	320.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	20.000 mm
Web Thickness	20.000 mm
Height of Center Web	850.000 mm



Number of Bolts in Baseplate

4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	65553.58	kgf
Maximum Transverse Saddle Shear Load	5618.06	kgf
Maximum Longitudinal Saddle Shear Load	11236.12	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	51394.27	kgf
Maximum Transverse Saddle Shear Load	837.25	kgf
Maximum Longitudinal Saddle Shear Load	277.90	kgf

Weights:

Fabricated - Bare w/o removable internals	64485.2	kg
Shop Test - Fabricated + water (full)	96897.3	kg
Shipping - Fab. + rem. intls.+ shipping app.	64485.2	kg
Erected - Fab. + rem. intls.+ insul. (etc)	64485.2	kg
Empty - Fab. + intls. + details + wghts.	64485.2	kg
Operating - Empty + operating liquid (no ca)	88874.8	kg
Field Test - Empty weight + test medium (full)	96897.3	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-Tube Inlet	18.5	...	...	OK	10.07	OK	Passed
N1-Shell Inlet	24	OK	...	OK	9.83	OK	Passed
N3-Shell Liquid	24	OK	...	OK	9.60	OK	Passed
N4-Shell Vapour	24	...	...	OK	4.92	OK	No Calc[*]
K1-Pressure Gau	24	...	...	OK	4.45	OK	No Calc[*]
N5-Tube Outlet	18.5	...	...	OK	9.94	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 24 Nozzle : N1-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 18.5 Nozzle : N2-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N2-Tube Inlet	1274.175	270.000	1536.000
40	N1-Shell Inlet	10334.175	90.000	717.200
40	N3-Shell Liquid	2764.175	270.000	412.400
40	N4-Shell Vapour	2764.175	90.000	93.325
40	K1-Pressure Gaug	9674.175	90.000	66.401
50	N5-Tube Outlet	11974.174	90.000	1536.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1-Pressure Gaug	1.000 in	XXS	LWN	1.315	9.093	...	...	318.15	300
N4-Shell Vapour	2.000 in	XXS	WNF	2.375	11.074	...	...	318.48	600
N3-Shell Liquid	8.000 in	Actual	WNF	9.260	16.000	...	...	340.65	300
N1-Shell Inlet	14.000 in	Actual	WNF	15.732	22.000	...	...	365.88	300
N2-Tube Inlet	30.000 in	Actual	WNF	31.260	16.000	1494.00	28.00	504.93	150
N5-Tube Outlet	30.000 in	Actual	WNF	31.260	16.000	1494.00	28.00	554.93	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1-Pressure	SA-333 6	18.000	10.000	...	...	...
N4-Shell Va	SA-333 6	18.000	10.000	...	...	...
N3-Shell Li	SA-333 6	18.000	10.000	...	...	...
N1-Shell In	SA-333 6	18.000	10.000	...	...	...
N2-Tube Inl	SA-333 6	14.000	12.000	12.000	28.000	...
N5-Tube Out	SA-333 6	14.000	10.000	12.000	28.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1-Pressure Gaug	967.417	90.0	300.00	0.00	Main Shell
N4-Shell Vapour	276.417	90.0	300.00	0.00	Main Shell
N3-Shell Liquid	276.417	270.0	300.00	0.00	Main Shell
N1-Shell Inlet	1033.417	90.0	300.00	0.00	Main Shell
N2-Tube Inlet	127.417	270.0	400.00	0.00	LHS Ch Shell
N5-Tube Outlet	1197.417	90.0	450.00	0.00	RHS Ch Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 128.0mm THK X 2100.0mm OD	SA-516 60
2	WELD NECK FLANGE: 143.0mm THK X 2100.0mm OD	SA-350 LF2
1	CYLINDER: 18.0mm THK X 1900.0mm ID X 200.0cm	SA-516 70
1	BLIND FLANGE: 143.0mm THK X 2100.0mm OD	SA-516 70
2	TUBESHEET: 1936.0mm X 60.0mm THK	SA-350 LF2
4760	TUBES: 8700.0mm X 19.0mm DIA X 2.0mm THK	SA-334 6
2	SADDLE: 415.0mm X 120 DEG	SA-516 70
2	GASKET: 1970.0mm OD X 1932.0mm ID	...
128	BODY FLANGE BOLTS: 24.0mm DIA	SA-320 L7
256	NUTS FOR BODY FLANGE BOLTS: 24.0mm DIA	...
1	CYLINDRICAL SEGMENT 18.0mm THK X 3050.8mm X 858.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 2918.1mm X 858.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 3050.8mm X 200.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 2918.1mm X 200.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Main Shell [8]		B	-11	-11	-29	0.995	18.000	1.00	Yes
Left Saddle [24]		B	-27	-27	-29	0.995	18.000	1.00	Yes
New Saddle [24]		B	-27	-27	-29	0.995	18.000	1.00	Yes
N1-Shell Inlet[1]		B	-11	-29	-29	0.971	18.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
N3-Shell Liqui[1]		B	-11	-29	-29	0.976	18.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
N4-Shell Vapou[1]		B	-8	-104	-29	0.050	9.690	1.00	Yes
Nozzle Flg [5]		A	-18	-104					
K1-Pressure Ga[1]		B	-29	-104	-29	0.028	7.957	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
Tubesheet: SS[13]		B	-27	-37	-29	0.821	18.000	1.00	Yes
Warmest MDMT:			-8	-11					
LHS CH Cover [11]		C	-16	-18		0.978	30.000	1.00	No
LHS Ch Flg [11]		!	-46	-46		0.560	18.000	1.00	No
RHS Ch Flg [11]		!	-46	-46		0.560	18.000	1.00	No
RHS Ch Cover [11]		B	2	0		0.978	27.500	1.00	No
LHS Ch Shell [8]		B	-11	-47	-29	0.474	18.000	0.85	No
RHS Ch Shell [8]		B	-11	-47	-29	0.474	18.000	0.85	No
N2-Tube Inlet [1]		B	-11	-47	-29	0.471	18.000	1.00	No
Nozzle Flg [5]		A	-18	-48					
N5-Tube Outlet[1]		B	-11	-48	-29	0.462	18.000	1.00	No
Nozzle Flg [5]		A	-18	-48					
Tubesheet: CS[14]		B	-11	-47	-29	0.473	18.000	1.00	No
Bolting [21]			-101						
Bolting [21]			-101						
Warmest MDMT:			2	0					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-29.0		-33.0		Fail			
Channel/Tube		0.0		-33.0		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.



UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	23.454	kgf/cm ²
Channel Side Design Pressure	10.197	kgf/cm ²
Shell Side Design Temperature	180.0	°C
Channel Side Design Temperature	80.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-2	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-33.0	°C
MDMT (CET), Tube Side	-33.0	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175

0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM	
Importance Factor for IS-1893	2.000	
Soil Factor	1.000	
Zone Number	2.000	
Percent Seismic for Hydrotest	0.000	
Period of Vibration (User defined) Sec	0.000	
Damping Factor	2.000	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS CH Cover
Distance "FROM" to "TO"	12 cm
Flange Outside Diameter	2100 mm
Element Thickness	120 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	128 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²



Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 60	
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²
Allowable Stress, Hydrotest	2018.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2090.3	kgf/cm ²
UCS-66 Chart Curve Designation	C	
External Pressure Chart Name	CS-2	
UNS Number	K02100	
Product Form	Plate	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	210	cm
Liquid Density	999.54	kg/m ³

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Flange	
Description	LHS Ch Flg	
Distance "FROM" to "TO"	15.1	cm
Flange Inside Diameter	1900	mm
Element Thickness	115	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	143	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF2 +IT	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2349.6	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03011	
Class / Thickness / Grade	1 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	190	cm
Liquid Density	999.54	kg/m ³

Element#	3/7	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	LHS Ch Shell	



Distance "FROM" to "TO"	200	cm
Inside Diameter	1900	mm
Element Thickness	18	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2477.8	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	190	cm
Liquid Density	999.54	kg/m ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2-Tube Inlet	
Dist. from "FROM" Node / Offset dist	100	cm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	470.06	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	30	
Detail Type	Weight	
Detail ID	baffle	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	0	mm

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	858	cm
Inside Diameter	1900	mm
Element Thickness	18	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	23.454	kgf/cm ²
Design Temperature Internal Pressure	180	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	180	°C



Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 40
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 100 cm
 Width of Saddle 415 mm
 Height of Saddle at Bottom 1800 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 625 mm
 Thickness of Wear Plate 20 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.30000001
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 40
 Detail Type Saddle
 Detail ID New Saddle
 Dist. from "FROM" Node / Offset dist 758 cm
 Width of Saddle 415 mm
 Height of Saddle at Bottom 1800 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 625 mm
 Thickness of Wear Plate 20 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.0
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 40
 Detail Type Liquid
 Detail ID Liquid: 40
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 190 cm
 Liquid Density 599.73 kg/m³

Element From Node 40
 Detail Type Nozzle
 Detail ID N1-Shell Inlet
 Dist. from "FROM" Node / Offset dist 800 cm
 Nozzle Diameter 14 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 302.4 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID N3-Shell Liquid
 Dist. from "FROM" Node / Offset dist 43 cm
 Nozzle Diameter 8 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 270.0



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 140.42 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID N4-Shell Vapour
 Dist. from "FROM" Node / Offset dist 43 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule XXS
 Nozzle Class 600
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.1739 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K1-Pressure Gaug
 Dist. from "FROM" Node / Offset dist 734 cm
 Nozzle Diameter 1 in.
 Nozzle Schedule XXS
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 3.6361 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element# 5/7
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description RHS Ch Shell
 Distance "FROM" to "TO" 200 cm
 Inside Diameter 1900 mm
 Element Thickness 18 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 18 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 10.197 kgf/cm²
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 190 cm
 Liquid Density 999.54 kg/m³

Element From Node 50
 Detail Type Nozzle
 Detail ID N5-Tube Outlet
 Dist. from "FROM" Node / Offset dist 100 cm
 Nozzle Diameter 30 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 90.0
 Blind Flange (Y/N) N



Weight of Nozzle (Used if > 0) 485.22 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element# 6/7
 Element From Node 60
 Element To Node 70
 Element Type Flange
 Description RHS Ch Flg
 Distance "FROM" to "TO" 15.1 cm
 Flange Inside Diameter 1900 mm
 Element Thickness 115 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 143 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 10.197 kgf/cm²
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-350 LF2 +IT
 Allowable Stress, Ambient 1407.1 kgf/cm²
 Allowable Stress, Operating 1407.1 kgf/cm²
 Allowable Stress, Hydrotest 2294.2 kgf/cm²
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 2349.6 kgf/cm²
 UCS-66 Chart Curve Designation Impact Tested
 External Pressure Chart Name CS-2
 UNS Number K03011
 Class / Thickness / Grade 1 ::
 Product Form Forgings
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 60
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 190 cm
 Liquid Density 999.54 kg/m³

Element# 7/7
 Element From Node 70
 Element To Node 80
 Element Type Flange
 Description RHS Ch Cover
 Distance "FROM" to "TO" 11 cm
 Flange Outside Diameter 2100 mm
 Element Thickness 110 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 143 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 10.197 kgf/cm²
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Allowable Stress, Ambient 1407.1 kgf/cm²
 Allowable Stress, Operating 1407.1 kgf/cm²
 Allowable Stress, Hydrotest 2386 kgf/cm²
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 2477.8 kgf/cm²
 UCS-66 Chart Curve Designation B

PV Elite 28 Licensee: SL User
FileName : 401-E-32001-AK
Input Echo:

Step: 1 11:00am Jul 4,2026



HEXAGON

External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	210 cm
Liquid Density	999.54 kg/m ³

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XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
LHS CH Cover		12.3175	...	12	...
LHS Ch Flg		27.4175	...	15.1	...
LHS Ch Shell		227.417	...	200	...
Main Shell		1091.42	...	858	...
RHS Ch Shell		1297.42	...	200	...
RHS Ch Flg		1312.83	...	15.1	...
RHS Ch Cover		1323.83	...	11	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

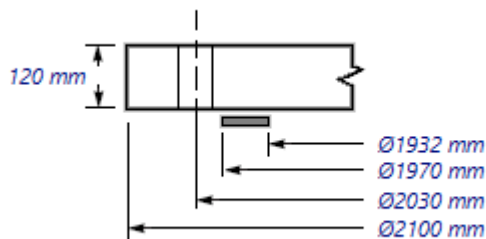
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Flange Input Data Values

Description: LHS CH Cover :

LHS CH Cover

Flange Type		Blind	
Design Pressure	P	10.41	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2100.000	mm
Flange Thickness	t	120.0000	mm
Flange Material		SA-516 60	
Flange Material UNS number		K02100	
Flange Allowable Stress At Temperature	Sfo	1203.20	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1203.20	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	6377.434	mm
Diameter of Bolt Circle	C	2030.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		64	
Flange Face Outside Diameter	Fod	1981.000	mm
Flange Face Inside Diameter	Fid	1900.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1970.000	mm
Gasket Inside Diameter	Gi	1932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1954.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1954^2 \cdot 10.4$
 $= 312236.812 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 7.77 \cdot \pi \cdot 1954 \cdot 3 \cdot 10.4 \\ &= 29779.393 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(312237 + 29779 + 0, 0) \\ &= 342016.188 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 703 \cdot 7.77 \cdot \pi \cdot 1954 + 0 \cdot 0 \cdot 0 \\ &= 335293.156 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa) \\ &= \max((342016 + 0 + 4 \cdot 0/1954)/1754, 335293/1754) \\ &= 195.012 \text{ cm}^2 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	195.012	200.159	
Radial Distance between Bolts and Edge:	29.000	35.000	
Circ. Spacing between the Bolts:	57.500	99.607	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= Ab \cdot Sa / (y \cdot \pi (Go + Gi)) \\ &= 200 \cdot 1754 / (703 \cdot \pi (1970 + 1932)) \\ &= 4.073 \text{ mm} \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned} &= Sa \cdot Ab \\ &= 1754 \cdot 200 \\ &= 351041.78 \text{ kgf} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= Wo \\ &= 342016.19 \text{ kgf} \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned} &= (C - G) / 2 \\ &= (2030 - 1954) / 2 \\ &= 37.7669 \text{ mm} \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned} &= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\ &= 1.9 \cdot 342016 \cdot 37.8 \cdot 1 / (117^2 \cdot 1954) + 0.3 \cdot 1 \cdot 10.4 \cdot 1954^2 / 117^2 \\ &= 962.98 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned} &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\ &= 1.9 \cdot 351042 \cdot 37.8 \cdot 1 / (117^2 \cdot 1954) \\ &= 94.15 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Operating [BSO]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 342016 / 200 \\ &= 1708.73 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (335293 / 200) \end{aligned}$$



= 1675.14 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	962.98	1203.20	94.15 1203.20
Bolting	1708.73	1753.82	1675.14 1753.82

Reqd. Blind Flange Thickness at Center 116.532 mm
 Reqd. Blind Flange Thickness at Gasket 38.499 mm

Estimated M.A.W.P. (Operating) 10.7 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 3221.3 kg
 Estimated Unfinished Weight of Forging at given Thk 3221.3 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.98, Temperature Reduction per Fig. UCS 66.1 = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -16 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	915.44	1203.20	89.50 1203.20
Bolting	1708.73	1753.82	1675.14 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 104.673 mm

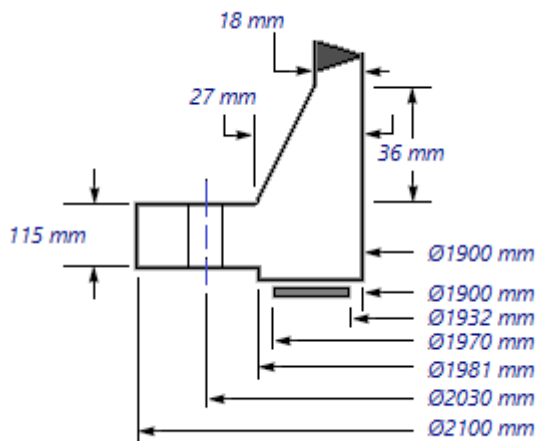
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Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	10.39	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1900.000	mm
Flange Outside Diameter	A	2100.000	mm
Flange Thickness	t	115.0000	mm
Thickness of Hub at Small End	go	18.0000	mm
Thickness of Hub at Large End	gl	27.0000	mm
Length of Hub	h	36.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material	SA-320 L7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	2030.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads	TEMA Metric		
Number of Bolts	64		
Flange Face Outside Diameter	Fod	1981.000	mm
Flange Face Inside Diameter	Fid	1900.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1970.000	mm
Gasket Inside Diameter	Gi	1932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot (1900/2 + 3)) / (1407 \cdot 0.85 - 0.6 \cdot 10.4) + Ca$$

= 11.3198 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 15) / (953 + 0.6 \cdot 15)$$

$$= 18.650 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 18) / (950 + 0.6 \cdot 18)$$

$$= 22.408 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc = T-ci	112.000 mm
Corroded Flange ID, Bcor = B+2*ci	1906.000 mm
Corroded Large Hub, g1Cor = g1-ci	24.000 mm
Corroded Small Hub, g0Cor = go-ci	15.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	38.000 mm

Gasket Contact Width, N = (Go - Gi)/2	19.000 mm
Basic Gasket Width, bo = N/2	9.500 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.767 mm
Gasket Reaction Diameter, G = Go - 2 * b	1954.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1954^2 \cdot 10.4$$

$$= 311637.000 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.77 \cdot \pi \cdot 1954 \cdot 3 \cdot 10.4$$

$$= 29722.188 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1906^2 \cdot 10.4$$

$$= 296372.906 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 311637 - 296373$$

$$= 15264.108 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(311637 + 29722 + 0, 0)$$

$$= 341359.188 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 7.77 \cdot \pi \cdot 1954 + 0 \cdot 0 \cdot 0$$

$$= 335293.188 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((341359 + 0 + 4 \cdot 0 / 1954) / 1754, 335293 / 1754)$$

$$= 194.638 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 24 + 6 \cdot 112 / (3 + 0.5)$$

$$= 240.000 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2030 \cdot \sin(\pi / 64)$$

$$= 99.607 \text{ mm}$$



ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{99.6}{(2 \cdot 24 + 112)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	194.638	200.159	
Radial Distance between Hub and Bolts:	33.500	38.000	
Radial Distance between Bolts and Edge:	29.000	35.000	
Circ. Spacing between the Bolts:	57.500	99.607	240.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 200 \cdot 1754 / (703 \cdot \pi(1970 + 1932))$$

$$= 4.073 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1754 \cdot 200$$

$$= 351041.78 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 341359 - 311637$$

$$= 29722.18 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2030 - 1954) / 2$$

$$= 37.7669 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2030 - 1906) / 2 + 37.8)$$

$$= 49.8834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2030 - 1906 - 24) / 2$$

$$= 50.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	296373.	50.0000	1.0000	14819.
Face Pressure, Mt	15264.	49.8834	1.0000	761.
Gasket Load, Mg	29722.	37.7669	1.0000	1123.
Gasket Seating, Matm	351042.	37.7669	1.0000	13258.

Total Moment for Operation, Mo 16703. kgf-m.
 Total Moment for Gasket seating, Mg 13258. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 169.086 mm
 Hub Ratio, h/h0 = HL / H0 0.213
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.600

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.393



Factor f per 4.16.5

1.653

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 2100/1906 \\ &= 1.102 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.1^2 \cdot \log_{10}(1.1))) / (1.1^2 - 1)) / (1.1 - 1) \\ &= 19.984 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.1^2(1 + 8.55246 \cdot \log_{10}(1.1)) - 1) / ((1.0472 + 1.9448 \cdot 1.1^2) \cdot (1.1 - 1)) \\ &= 1.877 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.1^2(1 + 8.55246 \cdot \log_{10}(1.1)) - 1) / (1.36136(1.1^2 - 1) \cdot (1.1 - 1)) \\ &= 21.961 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.1^2 + 1) / (1.1^2 - 1) \\ &= 10.349 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 22 \cdot 15^2 \cdot 169 / 0.39 \\ &= 2128445.000 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.9 / 169 \\ &= 0.005 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / (T + t^3/d) \\ &= (112 \cdot 0.0053 + 1) / (1.88 + 112^3/2128445) \\ &= 1.509 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.65 \cdot 16703 / 1906) / (1.51 \cdot 24^2) \\ &= 1666.32 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.65 \cdot 13258 / 1906) / (1.51 \cdot 24^2) \\ &= 1322.65 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (1.79 \cdot 16703 / 1906) / (1.51 \cdot 112^2) \\ &= 82.92 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (1.79 \cdot 13258 / 1906) / (1.51 \cdot 112^2) \\ &= 65.82 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (20 \cdot 16703 / (112^2 \cdot 1906)) - 10.3 \cdot 82.9 \\ &= 537.91 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (20 \cdot 13258 / (112^2 \cdot 1906)) - 10.3 \cdot 65.8$$

$$= 426.97 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (1666 + \max(82.9, 538)) / 2$$

$$= 1102.12 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1323 + \max(65.8, 427)) / 2$$

$$= 874.81 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 341359 / 200$$

$$= 1705.44 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (335293 / 200)$$

$$= 1675.14 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	1666.32	2110.70	1322.65 2110.70
Radial Flange	82.92	1407.13	65.82 1407.13
Tangential Flange	537.91	1407.13	426.97 1407.13
Maximum Average	1102.12	1407.13	874.81 1407.13
Bolting	1705.44	1753.82	1675.14 1753.82

Minimum Required Flange Thickness [Rigidity] 113.309 mm

Estimated M.A.W.P. (Operating) 10.6 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 597.9 kg
 Estimated Unfinished Weight of Forging at given Thk 735.3 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 13258 / 1 \cdot 100000 \cdot 0.39 / (1.47 \cdot 2067025 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.779 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 16703 / 1 \cdot 100000 \cdot 0.39 / (1.47 \cdot 2032634 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 13258 / 1 \cdot 100000 \cdot 0.39 / (1.51 \cdot 2067025 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.762 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 16703 / 1 \cdot 100000 \cdot 0.39 / (1.51 \cdot 2032634 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.976 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C



Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1419.09	2110.70	1095.75	2110.70
Radial Flange	89.91	1407.13	69.42	1407.13
Tangential Flange	421.78	1407.13	325.68	1407.13
Maximum Average	920.43	1407.13	710.71	1407.13
Bolting	1705.44	1753.82	1675.14	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 89.738 mm

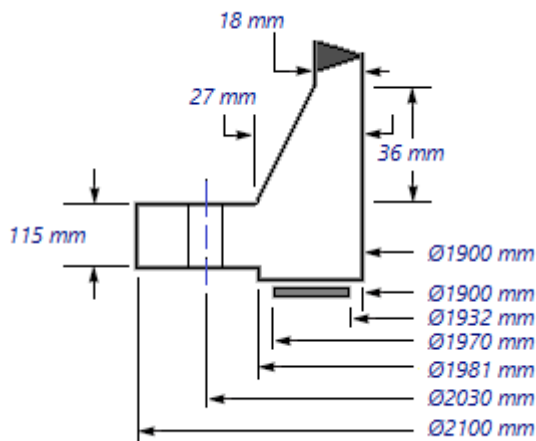
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Flange Input Data Values

Description: RHS Ch Flg :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	10.39	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1900.000	mm
Flange Outside Diameter	A	2100.000	mm
Flange Thickness	t	115.0000	mm
Thickness of Hub at Small End	go	18.0000	mm
Thickness of Hub at Large End	gl	27.0000	mm
Length of Hub	h	36.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material	SA-320 L7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	2030.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads	TEMA Metric		
Number of Bolts	64		
Flange Face Outside Diameter	Fod	1981.000	mm
Flange Face Inside Diameter	Fid	1900.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1970.000	mm
Gasket Inside Diameter	Gi	1932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot (1900/2 + 3)) / (1407 \cdot 0.85 - 0.6 \cdot 10.4) + Ca$$



= 11.3198 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 15) / (953 + 0.6 \cdot 15)$$

$$= 18.650 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 18) / (950 + 0.6 \cdot 18)$$

$$= 22.408 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc = T-ci	112.000 mm
Corroded Flange ID, Bcor = B+2*ci	1906.000 mm
Corroded Large Hub, g1Cor = g1-ci	24.000 mm
Corroded Small Hub, g0Cor = go-ci	15.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	38.000 mm

Gasket Contact Width, N = (Go - Gi)/2	19.000 mm
Basic Gasket Width, bo = N/2	9.500 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.767 mm
Gasket Reaction Diameter, G = Go - 2 * b	1954.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1954^2 \cdot 10.4$$

$$= 311637.000 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.77 \cdot \pi \cdot 1954 \cdot 3 \cdot 10.4$$

$$= 29722.188 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1906^2 \cdot 10.4$$

$$= 296372.906 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 311637 - 296373$$

$$= 15264.108 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(311637 + 29722 + 0, 0)$$

$$= 341359.188 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 7.77 \cdot \pi \cdot 1954 + 0 \cdot 0 \cdot 0$$

$$= 335293.188 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((341359 + 0 + 4 \cdot 0 / 1954) / 1754, 335293 / 1754)$$

$$= 194.638 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 24 + 6 \cdot 112 / (3 + 0.5)$$

$$= 240.000 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2030 \cdot \sin(\pi / 64)$$

$$= 99.607 \text{ mm}$$



ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{99.6/(2 \cdot 24 + 112)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	194.638	200.159	
Radial Distance between Hub and Bolts:	33.500	38.000	
Radial Distance between Bolts and Edge:	29.000	35.000	
Circ. Spacing between the Bolts:	57.500	99.607	240.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 200 \cdot 1754 / (703 \cdot \pi(1970 + 1932))$$

$$= 4.073 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1754(195 + 200) / 2$$

$$= 346200.50 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 341359 - 311637$$

$$= 29722.18 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2030 - 1954) / 2$$

$$= 37.7669 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2030 - 1906) / 2 + 37.8)$$

$$= 49.8834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2030 - 1906 - 24) / 2$$

$$= 50.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	296373.	50.0000	1.0000	14819.
Face Pressure, Mt	15264.	49.8834	1.0000	761.
Gasket Load, Mg	29722.	37.7669	1.0000	1123.
Gasket Seating, Matm	346200.	37.7669	1.0000	13075.

Total Moment for Operation, Mo 16703. kgf-m.
 Total Moment for Gasket seating, Mg 13075. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 169.086 mm
 Hub Ratio, h/h0 = HL / H0 0.213
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.600

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.393

Factor f per 4.16.5

1.653

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 2100/1906 \\ &= 1.102 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.1^2 \cdot \log_{10}(1.1))) / (1.1^2 - 1)) / (1.1 - 1) \\ &= 19.984 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.1^2(1 + 8.55246 \cdot \log_{10}(1.1)) - 1) / ((1.0472 + 1.9448 \cdot 1.1^2) \cdot (1.1 - 1)) \\ &= 1.877 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.1^2(1 + 8.55246 \cdot \log_{10}(1.1)) - 1) / (1.36136(1.1^2 - 1) \cdot (1.1 - 1)) \\ &= 21.961 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.1^2 + 1) / (1.1^2 - 1) \\ &= 10.349 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 22 \cdot 15^2 \cdot 169 / 0.39 \\ &= 2128445.000 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.9 / 169 \\ &= 0.005 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / (T + t^3/d) \\ &= (112 \cdot 0.0053 + 1) / (1.88 + 112^3/2128445) \\ &= 1.509 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.65 \cdot 16703 / 1906) / (1.51 \cdot 24^2) \\ &= 1666.32 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.65 \cdot 13075 / 1906) / (1.51 \cdot 24^2) \\ &= 1304.41 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (1.79 \cdot 16703 / 1906) / (1.51 \cdot 112^2) \\ &= 82.92 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (1.79 \cdot 13075 / 1906) / (1.51 \cdot 112^2) \\ &= 64.91 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (20 \cdot 16703 / (112^2 \cdot 1906)) - 10.3 \cdot 82.9 \\ &= 537.91 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (20 \cdot 13075 / (112^2 \cdot 1906)) - 10.3 \cdot 64.9$$

$$= 421.08 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (1666 + \max(82.9, 538)) / 2$$

$$= 1102.12 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1304 + \max(64.9, 421)) / 2$$

$$= 862.75 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 341359 / 200$$

$$= 1705.44 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (335293 / 200)$$

$$= 1675.14 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	1666.32	2110.70	1304.41 2110.70
Radial Flange	82.92	1407.13	64.91 1407.13
Tangential Flange	537.91	1407.13	421.08 1407.13
Maximum Average	1102.12	1407.13	862.75 1407.13
Bolting	1705.44	1753.82	1675.14 1753.82

Minimum Required Flange Thickness [Rigidity] 113.309 mm

Estimated M.A.W.P. (Operating) 10.6 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 597.9 kg
 Estimated Unfinished Weight of Forging at given Thk 735.3 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 13075 / 1 \cdot 100000 \cdot 0.39 / (1.47 \cdot 2067025 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.769 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 16703 / 1 \cdot 100000 \cdot 0.39 / (1.47 \cdot 2032634 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 13075 / 1 \cdot 100000 \cdot 0.39 / (1.51 \cdot 2067025 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.751 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 16703 / 1 \cdot 100000 \cdot 0.39 / (1.51 \cdot 2032634 \cdot 15^2 \cdot 169 \cdot 0.3)$$

$$= 0.976 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C



Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1419.09	2110.70	1080.64	2110.70
Radial Flange	89.91	1407.13	68.47	1407.13
Tangential Flange	421.78	1407.13	321.18	1407.13
Maximum Average	920.43	1407.13	700.91	1407.13
Bolting	1705.44	1753.82	1675.14	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 89.738 mm

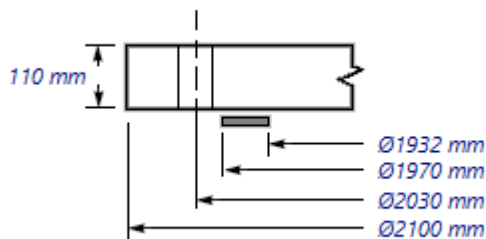
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Flange Input Data Values

Description: RHS Ch Cover :

RHS Ch Cover

Flange Type		Blind	
Design Pressure	P	10.41	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2100.000	mm
Flange Thickness	t	110.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	6377.434	mm
Diameter of Bolt Circle	C	2030.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		64	
Flange Face Outside Diameter	Fod	1981.000	mm
Flange Face Inside Diameter	Fid	1900.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1970.000	mm
Gasket Inside Diameter	Gi	1932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1954.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1954^2 \cdot 10.4$
 $= 312236.812 \text{ kgf}$



Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 7.77 \cdot \pi \cdot 1954 \cdot 3 \cdot 10.4 \\ &= 29779.393 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(312237 + 29779 + 0, 0) \\ &= 342016.188 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 703 \cdot 7.77 \cdot \pi \cdot 1954 + 0 \cdot 0 \cdot 0 \\ &= 335293.156 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa) \\ &= \max((342016 + 0 + 4 \cdot 0 / 1954) / 1754, 335293 / 1754) \\ &= 195.012 \text{ cm}^2 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	195.012	200.159	
Radial Distance between Bolts and Edge:	29.000	35.000	
Circ. Spacing between the Bolts:	57.500	99.607	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= Ab \cdot Sa / (y \cdot \pi (Go + Gi)) \\ &= 200 \cdot 1754 / (703 \cdot \pi (1970 + 1932)) \\ &= 4.073 \text{ mm} \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned} &= Sa(Am + Ab) / 2 \\ &= 1754 (195 + 200) / 2 \\ &= 346528.97 \text{ kgf} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= Wo \\ &= 342016.19 \text{ kgf} \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned} &= (C - G) / 2 \\ &= (2030 - 1954) / 2 \\ &= 37.7669 \text{ mm} \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned} &= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\ &= 1.9 \cdot 342016 \cdot 37.8 \cdot 1 / (107^2 \cdot 1954) + 0.3 \cdot 1 \cdot 10.4 \cdot 1954^2 / 107^2 \\ &= 1151.39 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned} &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\ &= 1.9 \cdot 346529 \cdot 37.8 \cdot 1 / (107^2 \cdot 1954) \\ &= 111.12 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Operating [BSO]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 342016 / 200 \\ &= 1708.73 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (335293 / 200) \end{aligned}$$



= 1675.14 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1151.39	1407.13	111.12 1407.13
Bolting	1708.73	1753.82	1675.14 1753.82

Reqd. Blind Flange Thickness at Center 107.983 mm
 Reqd. Blind Flange Thickness at Gasket 35.615 mm

Estimated M.A.W.P. (Operating) 10.7 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 2952.9 kg
 Estimated Unfinished Weight of Forging at given Thk 2952.9 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.98, Temperature Reduction per Fig. UCS 66.1 = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1089.44	1407.13	105.15 1407.13
Bolting	1708.73	1753.82	1675.14 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 96.799 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
LHS CH Cover		10.407	128	3	2100	1022.7	1
LHS Ch Flg		10.387	143	3	1900	1196.1	2
LHS Ch Shell		10.388	18	3	1900	1196.1	3
Main Shell		23.568	18	1.5	1900	1407.1	4
RHS Ch Shell		10.388	18	3	1900	1196.1	5
RHS Ch Flg		10.387	143	3	1900	1196.1	6
RHS Ch Cover		10.407	143	3	2100	1196.1	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
LHS CH Cover		10.1974	10.5	10.7	120	116.532	1
LHS Ch Flg		10.1974	10.4	10.7	115	113.309	2
LHS Ch Shell		10.1974	18.5	22.4	18	11.32	3
Main Shell		23.454	24	26.4	18	17.5984	4
RHS Ch Shell		10.1974	18.5	22.4	18	11.32	5
RHS Ch Flg		10.1974	10.4	10.7	115	113.309	6
RHS Ch Cover		10.1974	10.5	10.7	110	107.983	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

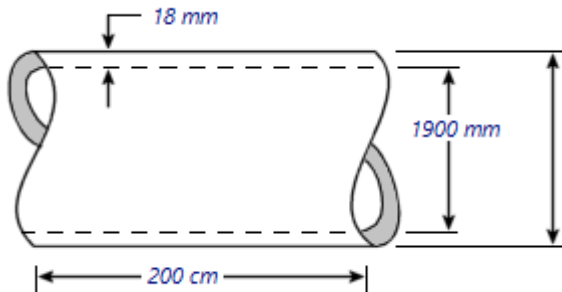
Shell Side MAWP = 24 kgf/cm²
 Shell Side MAPnc = 26.4 kgf/cm²
 Channel Side MAWP = 10.4 kgf/cm²
 Channel Side MAPnc = 10.7 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:				
Design Pressure	P	10.388 kgf/cm²	Int. Design Temperature	80 °C
Material		SA-516 70	Ext. Design Temperature	80 °C
Length		200.000 cm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm²
Inside Diameter	Di	1900.000 mm	Actual Stress	783.8 kgf/cm²
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area		121642.5 cm²	SG of Contents	1.000
Element Volume		5670575.0 cm³	Weight of Contents	5668.0 kgf
Empty Weight		1681.2 kgf	Operating Weight	7349.2 kgf
			Comp. All. Ope. Stress	1260.7 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot 953) / (1407 \cdot 0.85 - 0.6 \cdot 10.4)$$

$$= 8.3200 + 3.0000 = 11.3200 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.190 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 15) / (953 + 0.6 \cdot 15)$$

$$= 18.6 - 0.19 = 18.5 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 18) / (950 + 0.6 \cdot 18)$$

$$= 22.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.4 \cdot (953 + 0.6 \cdot 15)) / (0.85 \cdot 15)$$

$$= 783.756 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.938 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18, tr = 8.37, c = 3 mm, E* = 0.85



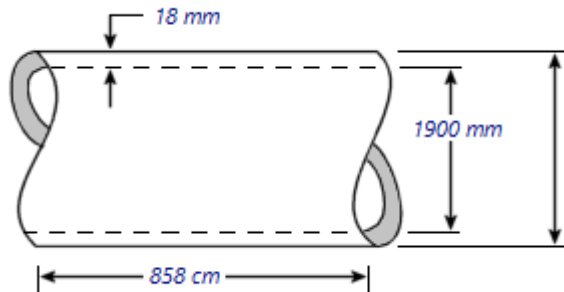
Thickness Ratio = $tr * E^* / (tg - c) = 0.47$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.449 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.474	Temperature Difference	36 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	23.568 kgf/cm²	Int. Design Temperature	180 °C	
Material		SA-516 70	Ext. Design Temperature	180 °C	
Length		858.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1900.000 mm	Actual Stress	1373.2 kgf/cm²	
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		521846.2 cm²	SG of Contents	0.600	
Element Volume		12747227.0 cm³	Weight of Contents	7644.9 kgf	
Empty Weight		7212.4 kgf	Operating Weight	14857.3 kgf	
			Comp. All. Ope. Stress	1200.5 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.6 \cdot 952) / (1407 \cdot 1 - 0.6 \cdot 23.6)$$

$$= 16.0984 + 1.5000 = 17.5984 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.114 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 16.5) / (952 + 0.6 \cdot 16.5)$$

$$= 24.1 - 0.11 = 24 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 18) / (950 + 0.6 \cdot 18)$$

$$= 26.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.6 \cdot (952 + 0.6 \cdot 16.5)) / (1 \cdot 16.5)$$

$$= 1373.232 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.938 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18, tr = 16.4, c = 1.5 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	24.036 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.995	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-11 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 40 to Tsht Ext. Chart: CS-2 at 180 °C

Elastic Modulus from Chart: CS-2 at 180 °C: 1999551 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1936.00	8580.00	117.33	4.4318	0.0002177	217.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*218)/(3*117) = 2.47 \text{ kgf/cm}^2$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.653	1936.00	8580.00	166.13	4.4318	0.0001314	131.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*131)/(3*166) = 1.05 \text{ kgf/cm}^2$$

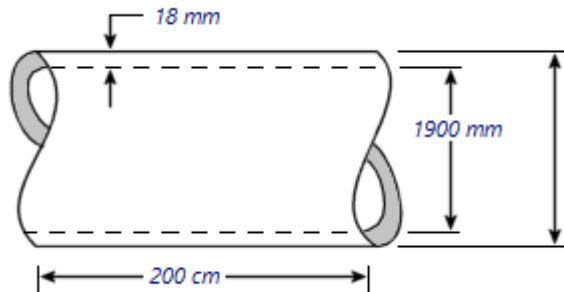
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1936.00	20374.44	117.33	10.5240	0.0000930	92.94

$$MAEP = (4*B)/(3*(Do/t)) = (4*92.9)/(3*117) = 1.06 \text{ kgf/cm}^2$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	10.388 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	80 °C	
Length		200.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1900.000 mm	Actual Stress	783.8 kgf/cm²	
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		121642.5 cm²	SG of Contents	1.000	
Element Volume		5670575.0 cm³	Weight of Contents	5668.0 kgf	
Empty Weight		1681.2 kgf	Operating Weight	7349.2 kgf	
			Comp. All. Ope. Stress	1200.5 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot 953) / (1407 \cdot 0.85 - 0.6 \cdot 10.4)$$

$$= 8.3200 + 3.0000 = 11.3200 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.190 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 15) / (953 + 0.6 \cdot 15)$$

$$= 18.6 - 0.19 = 18.5 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 0.85 \cdot 18) / (950 + 0.6 \cdot 18)$$

$$= 22.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.4 \cdot (953 + 0.6 \cdot 15)) / (0.85 \cdot 15)$$

$$= 783.756 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.938 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18, tr = 8.37, c = 3 mm, E* = 0.85



Thickness Ratio = $tr * E^* / (tg - c) = 0.47$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.449 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.474	Temperature Difference	36 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %



Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	31.247 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	30.490 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	34.060 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	26.439 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	33.539 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	33.539 kgf/cm ²
App 27-4	= M.A.W.P.	24.036 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	13.584 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.677 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	11.494 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	10.449 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	865.0	2386.0	0.363	13.77
Main Shell	1678.0	2386.0	0.703	31.44
RHS Ch Shell	865.0	2386.0	0.363	13.77

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-Tube Inlet	Nozzle	1203.20	1203.20	1.000
N2-Tube Inlet	Pad	1407.13	1407.13	1.000
N1-Shell Inlet	Nozzle	1203.20	1203.20	1.000
N3-Shell Liquid	Nozzle	1203.20	1203.20	1.000
N4-Shell Vapour	Nozzle	1203.20	1203.20	1.000
K1-Pressure Gaug	Nozzle	1203.20	1203.20	1.000
N5-Tube Outlet	Nozzle	1203.20	1203.20	1.000
N5-Tube Outlet	Pad	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS CH Cover	1203.20	1203.20	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Ch Shell	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
RHS Ch Shell	1407.13	1407.13	1.000
RHS Ch Flg	1407.13	1407.13	1.000
RHS Ch Cover	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------

Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2-Tube Inlet	336.26	1564.16	0.215
N1-Shell Inlet	272.93	1564.16	0.174
N3-Shell Liquid	218.48	1564.16	0.140
N4-Shell Vapour	73.05	1564.16	0.047
K1-Pressure Gaug	45.16	1564.16	0.029
N5-Tube Outlet	336.26	1564.16	0.215

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	31.247 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	30.490 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	34.060 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	26.439 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	33.539 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	33.539 kgf/cm ²
App 27-4	= M.A.W.P.	24.036 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	13.584 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.677 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	11.494 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	10.449 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	1046.2	2386.0	0.438	13.87
Main Shell	1995.7	2386.0	0.836	34.25
RHS Ch Shell	1046.2	2386.0	0.438	13.87

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-Tube Inlet	Nozzle	1203.20	1203.20	1.000
N2-Tube Inlet	Pad	1407.13	1407.13	1.000
N1-Shell Inlet	Nozzle	1203.20	1203.20	1.000
N3-Shell Liquid	Nozzle	1203.20	1203.20	1.000
N4-Shell Vapour	Nozzle	1203.20	1203.20	1.000
K1-Pressure Gaug	Nozzle	1203.20	1203.20	1.000
N5-Tube Outlet	Nozzle	1203.20	1203.20	1.000
N5-Tube Outlet	Pad	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS CH Cover	1203.20	1203.20	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Ch Shell	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
RHS Ch Shell	1407.13	1407.13	1.000
RHS Ch Flg	1407.13	1407.13	1.000
RHS Ch Cover	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
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Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2-Tube Inlet	338.52	1564.16	0.216
N1-Shell Inlet	297.36	1564.16	0.190
N3-Shell Liquid	238.04	1564.16	0.152
N4-Shell Vapour	79.58	1564.16	0.051
K1-Pressure Gaug	49.20	1564.16	0.031
N5-Tube Outlet	338.52	1564.16	0.216

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		34.06 kgf/cm ²
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1900.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		13.68 kgf/cm ²
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1900.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	4760
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	8700.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8580.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		1203.20 kgf/cm ²
Tube Allowable Stress At Ambient		1203.20 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2447.19 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type	Partial Strength	
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	918.900 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
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Tubesheet Material

SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	60.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1936.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

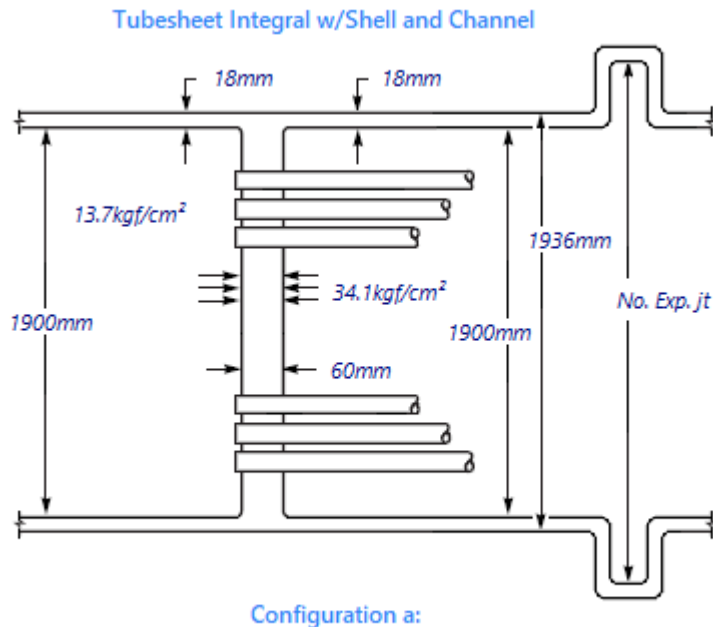
Unsupported Tube Span under consideration	l	760.000 mm
Tube End condition corresponding to Span (l)	k	0.80

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	
Shell Side Min. Design Pressure	Psd,min	-1.0546 kgf/cm²

Mean Shell Metal Temp. along Shell len.	T _{sm}	21.11 °C
Mean Tube Metal Temp. along Tube length	T _{tm}	21.11 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell	- TE-1	Carbon & Low Alloy Steels, Group 1
Shell	- TM-1	Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C



Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (14.7 \cdot (19/2 - 0)) / (1203 \cdot 1 + 0.4 \cdot 14.7)$
 $= 0.1157 + 0.0000 = 0.1157 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	21.11 °C
Elastic Modulus for Material			2038903.00 kgf/cm ²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	19.00	8580.00	9.50	50.0000	0.0121884	1251.46

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 181 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5643	19.00	8580.00	33.67	50.0000	0.0009702	860.23

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 860) / (3 \cdot 33.7) = 34.1 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5643 mm
Allowable Internal Pressure at Corroded thickness	276.60 kgf/cm ²
Required Internal Design Pressure	14.73 kgf/cm ²
Allowable External Pressure at Corroded thickness	181.22 kgf/cm ²
Required External Design Pressure	34.06 kgf/cm ²
Required Thickness due to Shell Side pressure	0.5643 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/60$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:
 $= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 2 (2067025/2067025) \cdot (1203/1407) \cdot (0); 19 - 2 \cdot 2)$
 $= 19.0000 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot r_o + dt$
 $= 2 \cdot 919 + 19$
 $= 1856.800 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (25 - 19) / 25$
 $= 0.240$

Equivalent outer tube limit radius [ao]:
 $= D_o / 2$
 $= 1857/2$
 $= 928.4000 \text{ mm}$

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$
 $= 25 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1857 \cdot 25) / (\pi \cdot 1857^2)) }$
 $= 25.0000 \text{ mm}$

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (25 - 19) / 25$
 $= 0.2400$

Ratio [ps]:
 $= a_s / a_o$
 $= 950/928$
 $= 1.0233$

Ratio [pc]:
 $= a_c / a_o$
 $= 950/928$
 $= 1.0233$

Compute parameter [xt]:
 $= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$
 $= 1 - 4760 ((19 - 2 \cdot 2) / (2 \cdot 928))^2$
 $= 0.6894$

Determine parameter [xs]:
 $= 1 - Nt(dt / (2 \cdot a_o))^2$
 $= 1 - 4760 (19 / (2 \cdot 928))^2$
 $= 0.5016$

Determine the value [h'g]:
 $= \max((h_g - CATC); 0)$ (For pressure only cases)
 $= \max((8 - 0); 0)$
 $= 8.000 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:
 $= \pi \cdot t_s(D_s + t_s) E_s / L$
 $= \pi \cdot 18 (1900 + 18) 2067025 / 8580$
 $= 261293.7656 \text{ kgf/mm}$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2 (19 - 2) 2067025 / 8580$$

$$= 257.3279 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 261294 / (4760 \cdot 257)$$

$$= 0.21332$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 261294 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1900 + 18) 18)$$

$$= 0.0098 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0098 \cdot 2067025 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 21600600.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1900 + 18) \cdot 21600600 / 60^3 \cdot (1 + 60 \cdot 0.0098 + (60 \cdot 0.0098)^2 / 2)$$

$$= 2024661.3750 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1900^2 / (4 \cdot 2067025 \cdot 18) \cdot (1 - 1900 \cdot 0.3 / (2(1900 + 18)))$$

$$= 0.20652\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + t_c) t_c)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1900 + 18) 18)$$

$$= 0.0098 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0098 \cdot 2067025 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 21600600.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1900 + 18 / \cos(0)) \cdot 21600600 / 60^3 \cdot (1 + 60 \cdot 0.0098 \cdot (1 + 1) / 2 + (60 \cdot 0.0098)^2 / 2 \cdot 1)$$

$$= 2024661.4 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1900^2 / (4 \cdot 2067025 \cdot 18) \cdot (1 - 1900 \cdot 0.3 / (2(1900 + 18)))$$

$$= 0.20652\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 2.400000 ; \mu^* = 0.240000 ; E^* = 421733 \text{ kgf/cm}^2$$

$$E^*/E = 0.204029 ; \nu^* = 0.407183$$

Note: As h/p (2.400) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:



$$= \left(\frac{24(1 - v^2) N t \cdot E t \cdot t t (d t - t t) a o^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.41^2) 4760 \cdot 2067025 \cdot 2}{(19 - 2) 928^2 (421734 \cdot 8580 \cdot 60^3)} \right)^{1/4}$$

$$= 9.2705$$

Values from Table 4.18.3

$$Z_d = 0.001846 ; Z_v = 0.011692 ; Z_m = 0.154073$$

$$Z_a = 0.561020E+04 ; Z_w = 0.011692$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1936/1857$$

$$= 1.0427$$

Compute coefficient [F]:

$$= (1 - v) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 421734 \cdot (2024661 + 2024661 + 2067025 \cdot \ln(1.04))$$

$$= 5.8134$$

Compute parameter [φ]:

$$= (1 + v) F$$

$$= (1 + 0.41) 5.81$$

$$= 8.1805$$

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.02 - 1 - 8.18 \cdot 0.012) / (1 + 8.18 \cdot 0.15)$$

$$= -0.032022$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.0018 + -0.032 \cdot 0.012) / 2 \cdot 9.27^4$$

$$= 5.4357$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.012 + -0.032 \cdot 0.15) / 2 \cdot 9.27^4$$

$$= 24.9589$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.012 + (1.02 - 1) 0.15) 9.27^4 / (1 + 8.18 \cdot 0.15)$$

$$= 49.9179$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 21600600 \cdot 0.0098 \cdot 0.020652205 (1 + 60 \cdot 0.0098)$$

$$= 7087.5435 \text{ mm}^2$$

Calculate parameter [w*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - w s$$

$$= 928^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 7088$$

$$= -6851.5498 \text{ mm}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 21600600 [0.020652205 (1 + 0.0098 \cdot 1 \cdot 60) - 0.000 (1/60 + 1 \cdot 0.0098/2)]$$

$$= 7087.5435 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1)/4 - (ps - 1)/2] - \omega c$$

$$= 928^2 [(1.02^2 + 1) \cdot (1.02 - 1)/4 - (1.02 - 1)/2] - 7088$$

$$= -6851.5503 \text{ mm}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 34.1 \{ 0.5 + 2(1 - 0.5)0.3 + [2/0.21 \cdot (1900/1857)^2]0.3 - [(1.02^2 - 1)/(1 \cdot 0.21)] - [(1 - 1)/(2 \cdot 1 \cdot 0.21)] [(0^2 - (1900)^2)/1857^2] \}$$

$$= 120.0638 \text{ kgf/cm}^2$$

Pressure [$P't$]:

$$= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] Pt$$

$$= [0.69 + 2(1 - 0.69)0.3 + 1/(1 \cdot 0.21)] 13.7$$

$$= 76.0899 \text{ kgf/cm}^2$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot y / (\pi \cdot ao^2)$$

$$= 4760 \cdot 257 \cdot 0 / (3.14 \cdot 928^2)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Pressure [P_w]:

$$= -yb \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 49.9 \cdot 0 / (2 \cdot \pi \cdot 928^2)$$

$$= -0.0000 \text{ kgf/cm}^2$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$$

$$= -(49.9/928^2) \cdot (-6852 \cdot 34.1 - -6852 \cdot 13.7)$$

$$= 8.0883 \text{ kgf/cm}^2$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)$$

$$= (49.9/928^2) \cdot (7088 \cdot 0 - 7088 \cdot 0)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.21 / (1 + 1 \cdot 0.21(5.44 + (1.02 - 1)25)) \cdot (120 - 76.1 + 0 + 0 + -0 + 8.09)$$

$$= 4.8638 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot yb / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-6852 \cdot 34.1 - -6852 \cdot 13.7) - (7088 \cdot 0 - 7088 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 8.18 \cdot 0.15)$$

$$= -617.8595 \text{ kgf}$$

Calculate factor [$Q3$]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.032 + 2 \cdot -618 / (4.86 \cdot 928^2)$$

$$= -0.06150$$



Fm Value from Table 4.18.3 = 0.030749

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.031 / 0.24) \cdot (2 \cdot 928 / (60 - 8))^2 \cdot 4.86 \\ &= 1191.8182 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta_T \cdot S_y = 1.5 \cdot 0.95 \cdot 2549 = 3633 \text{ kgf/cm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.023 / 0.24) \cdot (2 \cdot 928 / (32.3 - 8))^2 \cdot 4.24 \\ &= 3632.4678 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 32.3 + 0 + 0 \\ &= 32.2960 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta_T \cdot S_y) \\ &= (2 \cdot 0.24 \cdot 60 / 928) \cdot 1937 \\ &= 60.0990 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

$$\begin{aligned} \text{Tubesheet thickness} &= 4.8557 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 758.0417 \text{ kgf/cm}^2 \end{aligned}$$

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$\begin{aligned} &= \max(H_{reqB}, H_{reqS}) \\ &= \max(32.3, 4.86) \\ &= 32.2960 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.26, F_{tmax} = 4.16$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\ &= ((34.1 \cdot 0.5 - 13.7 \cdot 0.69) - (4.86) \cdot -0.26) / \\ &\quad (0.69 - 0.5) \\ &= 47.4599 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ &= ((34.1 \cdot 0.5 - 13.7 \cdot 0.69) - (4.86) \cdot 4.16) / \\ &\quad (0.69 - 0.5) \\ &= -67.0571 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 67.1 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 2324.8306 \text{ kgf/cm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.0018 + -0.061 \cdot 0.012)9.27^4); 1.25) \\ &= 1.2500 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ \frac{dt^2 + (dt - 2 \cdot tt)^2 }{ } }) / 4 \\ &= (\sqrt{ 19^2 + (19 - 2 \cdot 2)^2 }) / 4 \\ &= 6.0519 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ \frac{ 2 \cdot \pi^2 \cdot Et / S_{yt} }{ } } \\ &= \sqrt{ \frac{ 2 \cdot \pi^2 \cdot 2067025 / 2447 }{ } } \\ &= 135.0872 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0.8 \cdot 760 / 6.05 \\ &= 100.4650 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ &= (2447 / 1.25) \cdot (1 - 100 / (2 \cdot 135)) \\ &= 2324.831 \text{ kgf/cm}^2 \text{ (Never greater than } S_{ot} \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t, \max} \cdot \text{Tube Area} \\ &= 67.1 \cdot 1.07 \\ &= 71.63 \text{ kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19 - 2) 2 \\ &= 1.068 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2 (19 - 2) 2325 \\ &= 2483.245 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)} \\ &= 0.85 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 2325 \\ &= 2483.2446 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 2325 / 2325 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$\begin{aligned} &= F_d / F_t \\ &= 71.6 / 2483 \\ &= 0.029 \end{aligned}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)} \\ &= 0.55 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 2325 \\ &= 1634.1210 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 (\sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o }) \\ &= 2 (\sqrt{ ((0.75 \cdot 19)^2 + 1.07 \cdot 2 \cdot (19 - 2) 1 \cdot 0.029) - 0.75 \cdot 19 }) \end{aligned}$$



$$0.75 \cdot 19) \\ = 0.074 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 0.0368 mm
 Minimum Required Groove Weld Leg [agr]: 0.0368 mm

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 2651 = 3778 \text{ kgf/cm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 928^2 / ((1900 + 18) 18) \cdot [4.86 + (1.02^2 - 1) (34.1 - 13.7)] + 950^2 \cdot 13.7 / ((1900 + 18) 18)$
 $= 502.9082 \text{ kgf/cm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 18 mm, the minimum Shell length adjacent to the tubesheet should be 332.878 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	1191.8 <=	3632.5 kgf/cm ²	Pass
Tubesheet Shear Stress	156.8 <=	1937.4 kgf/cm ²	Pass
Maximum Tube Stress	67.1 <=	2324.8 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-67.1 <=	-2324.8 kgf/cm ²	Pass
Maximum Force on any one Tube	71.6 <=	2483.2 kgf	Pass
Axial Membrane Stress in Shell	502.9 <=	3777.8 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	32.296	60.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.037	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	0.037	2.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses		Case		Pass/	
			Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	37.660	...	1434	3633	258	1937	Ps+Pt-Th	D1 Ok
D2uc	49.615	...	2547	3633	402	1937	Ps+Pt-Th	D2 Ok
D3uc	32.296	...	1192	3633	157	1937	Ps+Pt-Th	D3 Ok
Max:	49.6155	... mm	0.701		0.208		(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Shell Stresses				:	Shell Band Stress				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	140	3777	...	...	:	...	...	...	...	Ok
D2uc	352	3777	...	...	:	...	...	...	...	Ok
D3uc	503	3777	...	...	:	...	...	...	...	Ok
Max RATIO	0.133				:	...				...

Tube, Shell and Channel Stress Summary:

Case#	Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	155	2324	-67	-2324	165	2483	...	3777	...	3777	Ok
D2uc	215	2324	-215	-2324	230	2483	...	3777	...	3777	Ok
D3uc	67	2324	-67	-2324	72	2483	...	3777	...	3777	Ok
Max RATIO	0.093		0.093		0.093		...		...		...

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	49.615	60.000	Ok
Tube thickness :	0.564	2.000	Ok
Tube-tubesheet fillet weld leg :	0.118	2.000	Ok
Tube-tubesheet groove weld leg :	0.118	2.000	Ok

Min Shell length of thk, (18.000) adj. to tubesheet: 332.878 mm

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.174 mm
 Due to Pressure + Thermal Shell Compresses by : -0.174 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0269	...	...	...	...	...
2	-0.1738	...	...	...	...	...
3	-0.1523	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	117	No Calc	No Calc
20	30	No Calc	...	112	No Calc	No Calc
30	Tsht	215.097	1936	15	0.00081767	804.227
Tsht	40	No Calc	...	15	No Calc	No Calc
40	Tsht	858	1936	16.5	0.00021774	217.693
Tsht	50	No Calc	...	16.5	No Calc	No Calc
50	60	215.103	1936	15	0.00081765	804.217
60	70	No Calc	...	112	No Calc	No Calc
70	80	No Calc	...	107	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	120	35.7287	...	No Calc
20	30	115	94.3864	...	No Calc
30	Tsht	18	No Calc	...	8.31
Tsht	40	18	No Calc	...	No Calc
40	Tsht	18	13.1532	1.05461	2.47
Tsht	50	18	No Calc	1.05461	No Calc
50	60	18	No Calc	...	8.31
60	70	115	92.6592	...	No Calc
70	80	110	32.9229	...	No Calc

Minimum

2.47

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	215.097	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	858	2037.44	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	215.103	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	3436.06	...	3436.06	...	343.606
20	30	734.283	510081	709.401	510727	73.4283
30	40	1681.21	5670575	1403.2	5706446	168.121
40	50	7212.4	12747227	6616.54	12824109	721.24
50	60	1681.21	5670575	1403.2	5706446	168.121
60	70	734.283	510081	709.401	510727	73.4283
70	80	3838.72	...	3838.72	...	383.872
Total		19318	25108540.00	18116	25258456.00	1931



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	...	6	...	...	Liquid: 10
20	Liqd	509.852	7.55	...	...	Liquid: 20
30	Liqd	5668.03	100	...	...	Liquid: 30
30	Nozl	517.07	100	-133.1	...	N2-Tube Inlet
30	Wght	299.995	100	...	...	baffle
40	Sadl	1020.58	100	135.6	...	Left Saddle
40	Sadl	1020.58	758	135.6	...	New Saddle
40	Liqd	7644.91	429	...	...	Liquid: 40
40	Nozl	332.641	800	112.78	...	N1-Shell Inlet
40	Nozl	154.457	43	-105.16	...	N3-Shell Liquid
40	Nozl	10.0913	43	96.9088	...	N4-Shell Vapour
40	Nozl	3.99971	734	95.7607	...	K1-Pressure Gaug
50	Liqd	5668.03	100	...	...	Liquid: 50
50	Nozl	533.74	100	133.1	...	N5-Tube Outlet
60	Liqd	509.852	7.55	...	...	Liquid: 60
70	Liqd	...	5.5	...	...	Liquid: 70
30	FTsh	815.447	203	...	...	
30	Tube	37711.2	635	...	...	Tubes
30	RTsh	815.447	1067	...	...	Tubesheet

Total Weight of Each Detail Type:

Saddles	2041.2
Liquid	20000.7
Nozzles	1552.0
Weights	300.0
Exchanger Components	39342.1
Liquid in Tubes	4388.9
Sum of the Detail Weights	67624.8 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	21250.0	21250.0	21250.0	21250.0	21250.0	21250.0
Saddles	2041.2	2041.2	2041.2	2041.2	2041.2	2041.2
Nozzles	1552.0	1552.0	1552.0	1552.0	1552.0	1552.0
Wld Weights	300.0	300.0	300.0	300.0	300.0	300.0
Exchanger	39342.1	39342.1	39342.1	39342.1	39342.1	39342.1
Ope. Liquid	...	...	...	...	...	20000.7



Tube Ope Lqd	...	...	...	...	...	4388.9
Test Liquid	...	25097.3	...	...	...	...
Tube Tst Lqd	...	7314.8	...	...	...	...
Totals	64485.2	96897.3	64485.2	64485.2	64485.2	88874.8

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	64485.2 kg
Shop Test Wt.	- Fabricated weight + water (full)	96897.3 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	64485.2 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	64485.2 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	64485.2 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	88874.8 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	87553.0 kg
Field Test Wt.	- Empty weight + test medium (full)	96897.3 kg

Exchanger Tube Data

Volume of Exchanger tubes :	7318105.0 cm ³
Weight of Ope Liq in tubes :	4388.9 kg
Weight of Water in tubes :	7314.8 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	43080.7	...
20	30	17927.3	10684.6
30	40	121642	119381
40	50	521846	512142
50	60	121642	119381
60	70	17927.3	10684.6
70	80	44070.3	...

Total	888136.6	772272.7 cm ²
Total	956.0	831.3 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	Max Pressure DNV
N2-Tube Inlet	18.62	19.99	80	150	GR 1.1	12.387	13.416	18.621	15.545
N1-Shell Inlet	45.20	52.01	180	300	GR 1.1	22.414	45.195	45.195	45.195
N3-Shell Liqui	45.20	52.01	180	300	GR 1.1	28.187	39.606	45.195	39.606
N4-Shell Vapou	90.35	104.01	180	600	GR 1.1	67.584	67.940	90.350	67.940
K1-Pressure Ga	45.20	52.01	180	300	GR 1.1	...	...	...	...
N5-Tube Outlet	18.62	19.99	80	150	GR 1.1	12.387	13.416	18.621	15.545

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 39.606 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 52.007 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 13.416 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 19.987 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N2-Tube Inlet	787.400	0.39	11468	2060	10.449	13.416	Pass
N1-Shell Inlet	389.001	0.50	2497	959	24.036	45.195	Pass
N3-Shell Liquid	248.539	0.50	815	551	24.036	39.606	Pass
N4-Shell Vapour	77.851	0.50	59	163	24.036	67.940	Pass
N5-Tube Outlet	787.400	0.39	11468	2060	10.449	13.416	Pass

Evaluating Flange on Nozzle [N2-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(11468) + 4 \cdot \text{abs}(2060) \cdot 787.400 \leq \pi \cdot 787.400^3 \cdot ((18.621 - 10.449) + .386 \cdot 18.621)$$

$$= 0.18997E+06 \leq 0.23547E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.621, 0.386 \cdot 18.621 + 18.621 - 189971.8 / (\pi \cdot 787.400^3))$$

$$= \min(18.621, 13.416)$$

$$= 13.416 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N1-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(2497) + 4 \cdot \text{abs}(959) \cdot 389.001 \leq \pi \cdot 389.001^3 \cdot ((45.195 - 24.036) + .500 \cdot 45.195)$$

$$= 0.41450E+05 \leq 0.80919E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.195, 0.500 \cdot 45.195 + 45.195 - 41450.3 / (\pi \cdot 389.001^3))$$

$$= \min(45.195, 45.378)$$

$$= 45.195 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N3-Shell Liquid]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(815) + 4 \cdot \text{abs}(551) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot ((45.195 - 24.036) + .500 \cdot 45.195)$$



= 0.13595E+05 ≤ 0.21105E+05 Check Passes

Maximum Pressure for Flange on Nozzle N3-Shell Liquid [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(45.195, 0.500 \cdot 45.195 + 45.195 - 13595.2 / (\pi \cdot 248.539^3)) \\ &= \min(45.195, 39.606) \\ &= 39.606 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [N4-Shell Vapour]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(59) + 4 \cdot \text{abs}(163) 77.851 \leq \pi \cdot 77.851^3 [(90.350 - 24.036) + 0.500 \cdot 90.350] \\ &= 0.10018E+04 \leq 0.16526E+04 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell Vapour [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(90.350, 0.500 \cdot 90.350 + 90.350 - 1001.8 / (\pi \cdot 77.851^3)) \\ &= \min(90.350, 67.940) \\ &= 67.940 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [N5-Tube Outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(11468) + 4 \cdot \text{abs}(2060) 787.400 \leq \pi \cdot 787.400^3 [(18.621 - 10.449) + 0.386 \cdot 18.621] \\ &= 0.18997E+06 \leq 0.23547E+06 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N5-Tube Outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(18.621, 0.386 \cdot 18.621 + 18.621 - 189971.8 / (\pi \cdot 787.400^3)) \\ &= \min(18.621, 13.416) \\ &= 13.416 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 6.897

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.89 • 1 • 0.6 • 57.5²)

= 1770.3 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.3 • 0.84 • 1770

= 448.6 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	2.10	1.0000	1.0000	1.0000	57.50	0.8379
20	2.10	1.0000	1.0000	1.0000	57.50	0.8379
30	2.10	1.0000	1.0000	1.0000	57.50	0.8379
40	2.10	1.0000	1.0000	1.0000	57.50	0.8379
50	2.10	1.0000	1.0000	1.0000	57.50	0.8379
60	2.10	1.0000	1.0000	1.0000	57.50	0.8379
70	2.10	1.0000	1.0000	1.0000	57.50	0.8379

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
baffle	0.00	210.00	180.60	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	210	252	3024	180.598	50.3197
20	30	210	228	3442.8	180.598	57.2886
30	40	210	232.32	46464	180.598	773.166
40	50	210	232.32	199331	180.598	3316.88



50	60	210	232.32	46464	180.598	773.166
60	70	210	228	3442.8	180.598	57.2886
70	80	210	252	2772	180.598	46.1264

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1.4) \cdot 1$
 $= 0.131$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 9501 \cdot 0.13 \cdot 1$
 $= 1248.5 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.13 \cdot 85512 \cdot 1$
 $= 7490.7 \text{ kgf}$

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------



From	To	Height cm	Weight kgf	Ope Load kgf
10	20	105	9501.31	1248.46
20	30	95	9501.31	1248.46
30	40	95	9501.31	1248.46
40	Sad1	95	9501.31	1248.46
Sad1	50	95	9501.31	1248.46
40	50	95	9501.31	1248.46
50	60	95	9501.31	1248.46
60	70	95	9501.31	1248.46
70	80	105	9501.31	1248.46

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	662.418 cm
Center of Gravity of the Liquid	662.426 cm
Center of Gravity of the Nozzles	707.542 cm
Center of Gravity of the Added Weights (Operating)	127.417 cm
Center of Gravity of the Added Weights (Empty)	127.417 cm
Center of Gravity of the Tubesheet(s)	662.418 cm
Center of Gravity of the Tubes	662.417 cm
Center of Gravity of Bare Shell New and Cold	676.069 cm
Center of Gravity of Bare Shell Corroded	676.974 cm
Vessel CG in the Operating Condition	664.699 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	665.513 cm
Vessel CG in the Test Condition	664.479 cm

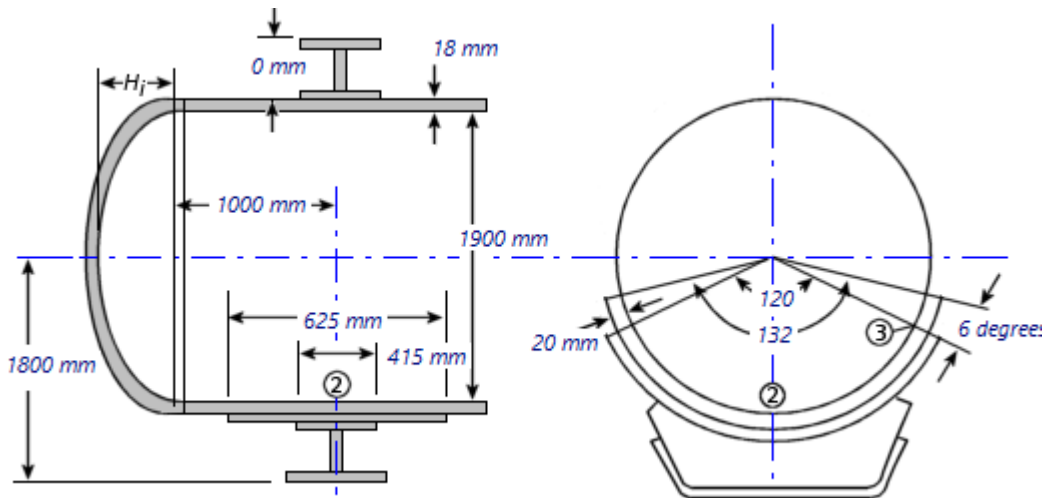
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	959.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	858.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	415.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	63925.75	kgf
Pressure used in Analysis	P	23.511 kgf/cm ²

Horizontal Vessel Analysis Results:		
	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	523.30	1407.13
Long. Stress at Bottom of Midspan	844.26	1407.13
Long. Stress at Top of Saddles	762.73	1407.13
Long. Stress at Bottom of Saddles	640.02	1407.13
Tangential Shear in Shell	362.42	1125.71
Circ. Stress at Horn of Saddle	452.02	1758.92
Circ. Compressive Stress in Shell	21.78	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (5074/2 + 0) \cdot 1800/1711$$

$$= 8006.0 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(842, 0) \cdot 1800/6580$$

$$= 230.4 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(11236, 0) \cdot 1800/6580$$

$$= 3073.7 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (11236/2 + 0) \cdot 1800/1711$$

$$= 17728.1 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 46198 + \max(230, 8006, 3074, 17728)$$

$$= 63925.8 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1771 (0 + \pi/4(1.94 \cdot 1.2)^2)$$

$$= 8258.274 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	64946.33	kgf
Transverse Shear Load	5618.06	kgf
Longitudinal Shear Load	11236.12	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1900/2 + 18 + 20) \sin(120/2), 2 \cdot 960)$$

$$= 1711.266 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L} \cdot \text{a})) / (1 + (4\text{hm})/3\text{L})]$$

$$= -63926 \cdot 100 [1 - (1 - 100/858 + (96^2 - 0^2)/(\text{L} \cdot \text{a})) / (1 + (4 \cdot 0)/(3 \cdot 858))]$$

$$= -4019.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot \text{L} / 4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4\text{a}/\text{L})$$

$$= 63926 \cdot 858 / 4 ((1 + 2(96^2 - 0^2)/(858^2)) / (1 + (4 \cdot 0)/(3 \cdot 858)) - 4 \cdot 100/858)$$

$$= 76626.5 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 23.5 \cdot 960/(2 \cdot 16.5) - 76626/(\pi \cdot 960^2 \cdot 16.5)$$

$$= 523.30 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 23.5 \cdot 960 / (2 \cdot 16.5) + 76626 / (\pi \cdot 960^2 \cdot 16.5)$$

$$= 844.26 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.5 \cdot 960 / (2 \cdot 16.5) - 4019 / (0.11 \cdot \pi \cdot 960^2 \cdot 16.5)$$

$$= 762.73 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.5 \cdot 960 / (2 \cdot 16.5) + 4019 / (0.19 \cdot \pi \cdot 960^2 \cdot 16.5)$$

$$= 640.02 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 63926 (858 - 2 \cdot 100) / (858 + (4 \cdot 0 / 3))$$

$$= 49024.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 49025 / (960 \cdot 16.5)$$

$$= 362.42 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 16.5)}$$

$$= 98.156 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(415 + 1.56 \cdot \sqrt{(960 \cdot 16.5)}, 2 \cdot 100)$$

$$= 611.31 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407 / 1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot tr))$$

$$= -0.76 \cdot 63926 \cdot 0.1 / (611 (16.5 + 1 \cdot 20))$$

$$= -21.78 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -63926 / (4(16.5 + 1 \cdot 20) 611) -$$

$$3 \cdot 0.053 \cdot 63926 / (2(16.5 + 1 \cdot 20)^2)$$

$$= -452.02 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 6580.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12620 \text{E-}04 \cdot 6580 (180 - 21.1)$$

$$= 13.194 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	320.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	850.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(952 \cdot 16.5)} + 625)^3) / 12$$

$$= 30.714 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 30.7 + 135 (289 - 8.25)^2$$

$$= 106631.727 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	820.5	16.5	8.3	135.4	111685.9	0.107E+06
Wearplate	625.0	20.0	26.5	125.0	331250.0	0.861E+05
Web	20.0	784.0	428.5	156.8	6718880.0	0.111E+06
BasePlate	320.0	28.0	834.5	89.6	7477121.0	0.267E+06
Totals	...	...	...	506.8	14638937.0	0.570E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14638937 / 507$$

$$= 288.864 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 63926$$

$$= 13010.2754 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 35.0304 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1391.2390 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1800 - 952 \cdot \sin(120/2) / 1.05$$

$$= 1013.116 \text{ mm}$$

Bending Moment, M = Fh * d = 13180.9307 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 66.7489 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1545.8213 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(63926 + 1021)320) / (4 \cdot 1800 \cdot 1546)}$$

$$= 23.668 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (950 + 18 + 20)$$

$$= 1711.266 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1711 / (4 - 1)$$

$$= 570.422 \text{ mm}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 570 \cdot 320 / 2$$

$$= 912.675 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 64946 / (1800 \cdot 320)$$

$$= 11.275 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 913 \cdot 11.3$$

$$= 10290.784 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 79 + 57$$

$$= 136.042 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 10291 / 136$$

$$= 75.644 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	415.0	...	83.0	...	0.119E+05

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 415 - 20$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 136$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 415 / 2$$

$$= 207.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (11912 / 136) }$$

$$= 93.575 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1800 - \cos(120 / 2) (950 + 18 + 20) - 28$$

$$= 1278.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 1973951 / 2319) }$$

$$= 129.631$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1278 / 93.6$$

$$= 13.657$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 11236 / (2 \cdot 1800) \cdot 570 \cdot 1278 / 2$$

$$= 1137.658 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.7 < 130$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.7)^2 / (2 \cdot 130^2)) 2319 / (5/3 + 3 \cdot (13.7) / (8 \cdot 130) - (13.7^3) / (8 \cdot 130^3))$$

$$= 1352 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 75.6/1352 + (1138 \cdot 208/119122272) / 1546$$

$$= 0.184$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	395.0	0.0	79.0	0.0	0.119E+05
Web	570.4	20.0	0.0	114.1	0.0	38.0
Totals	...	...	...	193.1	...	0.119E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/193$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 415/2$$

$$= 207.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1800 - \sqrt{ (988 + 20)^2 - (570/2)^2 } - 28$$

$$= 826.062 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 11949/193 }$$

$$= 78.667 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 826/78.7$$

$$= 10.501$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 11236 / (2 \cdot 1800)$$

$$= 3.121 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.12 \cdot 570 \cdot 826$$

$$= 1470.698 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1471/575852$$

$$= 255.395 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($10.5 < 130$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.5)^2 / (2 \cdot 130^2)) 2319 / (5/3 + 3 \cdot (10.5) / (8 \cdot 130) - (10.5^3) / (8 \cdot 130^3))$$

$$= 1362 \text{ kgf/cm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 122/1362 + (1471 \cdot 208/11949)/1546$$

$$= 0.255$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	47218.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3073.7 kgf
Maximum Transverse Force	Ft	5618.1 kgf
Maximum Longitudinal Force	F1	11236.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (Bolt Area \cdot Number of Bolts)$$

$$= 11236 / (6.46 \cdot 4)$$

$$= 435.0 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (Bolt Area \cdot Number of Bolts)$$

$$= 5618 / (6.46 \cdot 4)$$

$$= 217.5 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (BoltShearAllowable \cdot Nbolts)$$

$$= 11236 / (1100 \cdot 4)$$

$$= 2.5537 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1800 \cdot 5618 + 0$$

$$= 10112.52 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 10113 / 47218$$

$$= 214.17 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 2.55, 0]$$

$$= 2.5537 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 18, tr = 16.4, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	959.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	858.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	415.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		64533.00	kgf
Pressure used in Analysis	P	23.511	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		521.77	1407.13
Long. Stress at Bottom of Midspan		845.79	1407.13
Long. Stress at Top of Saddles		763.48	1407.13
Long. Stress at Bottom of Saddles		639.60	1407.13
Tangential Shear in Shell		365.87	1125.71
Circ. Stress at Horn of Saddle		456.32	1758.92
Circ. Compressive Stress in Shell		21.99	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (5074/2 + 0) \cdot 1800/1711$
 $= 8006.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(842, 0) \cdot 1800/6580$
 $= 230.4 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(11236, 0) \cdot 1800/6580$
 $= 3073.7 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (11236/2 + 0) \cdot 1800/1711$
 $= 17728.1 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:



$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 46805 + \max(230, 8006, 3074, 17728) \\
 &= 64533.0 \text{ kgf}
 \end{aligned}$$

Longitudinal Wind Force [F_L]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 1771 (0 + \pi/4(1.94 \cdot 1.2)^2) \\
 &= 8258.274 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	65553.58	kgf
Transverse Shear Load	5618.06	kgf
Longitudinal Shear Load	11236.12	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(1900/2 + 18 + 20)\sin(120/2), 2 \cdot 960) \\
 &= 1711.266 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -64533 \cdot 100 [1 - (1 - 100/858 + (96^2 - 0^2)/(2 \cdot 100 \cdot 858))/(1 + (4 \cdot 0)/(3 \cdot 858))] \\
 &= -4057.3 \text{ kgf-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 \{ (1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L \} \\
 &= 64533 \cdot 858/4 \{ (1 + 2(96^2 - 0^2)/(858^2))/(1 + (4 \cdot 0)/(3 \cdot 858)) - 4 \cdot 100/858 \} \\
 &= 77354.4 \text{ kgf-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 23.5 \cdot 960/(2 \cdot 16.5) - 77354/(\pi \cdot 960^2 \cdot 16.5) \\
 &= 521.77 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 23.5 \cdot 960/(2 \cdot 16.5) + 77354/(\pi \cdot 960^2 \cdot 16.5) \\
 &= 845.79 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 23.5 \cdot 960/(2 \cdot 16.5) - 4057/(0.11 \cdot \pi \cdot 960^2 \cdot 16.5) \\
 &= 763.48 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 23.5 \cdot 960/(2 \cdot 16.5) + 4057/(0.19 \cdot \pi \cdot 960^2 \cdot 16.5) \\
 &= 639.60 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 64533 (858 - 2 \cdot 100)/(858 + (4 \cdot 0/3)) \\
 &= 49490.3 \text{ kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ₂]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 49490 / (960 \cdot 16.5)$$

$$= 365.87 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 16.5)}$$

$$= 98.156 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(415 + 1.56 \cdot \sqrt{(960 \cdot 16.5)}, 2 \cdot 100)$$

$$= 611.31 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 64533 \cdot 0.1 / (611(16.5 + 1 \cdot 20))$$

$$= -21.99 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -64533 / (4(16.5 + 1 \cdot 20)611) - 3 \cdot 0.053 \cdot 64533 / (2(16.5 + 1 \cdot 20)^2)$$

$$= -456.32 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	320.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	850.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(952 \cdot 16.5)} + 625)16.5^3) / 12$$

$$= 30.714 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 30.7 + 135(289 - 8.25)^2$$

$$= 106631.727 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	820.5	16.5	8.3	135.4	111685.9	0.107E+06
Wearplate	625.0	20.0	26.5	125.0	331250.0	0.861E+05
Web	20.0	784.0	428.5	156.8	6718880.0	0.111E+06
BasePlate	320.0	28.0	834.5	89.6	7477121.0	0.267E+06
Totals	...	...	...	506.8	14638937.0	0.570E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14638937/507$$

$$= 288.864 \text{ mm}$$

Angle [β]:



$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 64533 \\ &= 13133.8633 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 35.3632 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1800 - 952 \cdot \sin(120/2) / 1.05 \\ &= 1013.116 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 13306.1396 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 67.3829 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\ &= \sqrt{(3(64533 + 1021) \cdot 320) / (4 \cdot 1800 \cdot 1546)} \\ &= 23.779 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (950 + 18 + 20) \\ &= 1711.266 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (N_{\text{ribs}} - 1) \\ &= 1711 / (4 - 1) \\ &= 570.422 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 570 \cdot 320 / 2 \\ &= 912.675 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 65554 / (1800 \cdot 320) \\ &= 11.381 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 913 \cdot 11.4 \\ &= 10387.004 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 79 + 57 \\ &= 136.042 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 10387/136 \\ &= 76.351 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	415.0	...	83.0	...	0.119E+05

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 415 - 20
 = 395.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/136
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 415/2
 = 207.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{11912/136}$
 = 93.575 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1800 - cos(120/2)(950 + 18 + 20) - 28
 = 1278.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2038903/2319)}$
 = 131.746

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1278/93.6
 = 13.657

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 11236 / (2 • 1800) • 570 • 1278/2
 = 1137.658 kgf-m.

Compressive Allowable, KL/r < Cc (13.7 < 132) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (13.7)^2 / (2 \cdot 132^2)) 2319 / (5/3 + 3 \cdot (13.7) / (8 \cdot 132) - (13.7)^3 / (8 \cdot 132^3))$
 = 1352 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 76.4/1352 + (1138 • 208/119122272) / 1546
 = 0.185

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	395.0	0.0	79.0	0.0	0.119E+05
Web	570.4	20.0	0.0	114.1	0.0	38.0
Totals	...	...	...	193.1	...	0.119E+05

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/193



= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 415/2
 = 207.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
 = 1800 - $\sqrt{(988 + 20)^2 - (570/2)^2}$ - 28
 = 826.062 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{11949/193}$
 = 78.667 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 826/78.7
 = 10.501

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 11236/(2 • 1800)
 = 3.121 kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 3.12 • 570 • 826
 = 1470.698 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 1471/575852
 = 255.395 kgf/cm²

Compressive Allowable, KL/r < Cc (10.5 < 132) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$
 = $(1 - (10.5)^2 / (2 \cdot 132^2)) 2319 /$
 $(5/3 + 3 \cdot (10.5) / (8 \cdot 132) - (10.5^3) / (8 \cdot 132^3))$
 = 1362 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 123/1362 + (1471 • 208/11949)/1546
 = 0.255

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	47825.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3073.7 kgf
Maximum Transverse Force	Ft	5618.1 kgf
Maximum Longitudinal Force	Fl	11236.1 kgf
Saddle Bolted to Steel Foundation	No	

Friction force due to thermal growth, [Ff]:

= 14165.5 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:



= max(Ff, FI) / (Bolt Area • Number of Bolts)
= 14165/(6.46 • 4)
= 548.4 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= Ft / (Bolt Area • Number of Bolts)
= 5618/(6.46 • 4)
= 217.5 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 11236/(1100 • 4)
= 2.5537 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1800 • 5618 + 0
= 10112.52 kgf-m.

Eccentricity (e):
= Rmom / QO
= 10113/47825
= 211.45 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 2.55, 0]
= 2.5537 cm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 18, tr = 16.4, c = 1.5 mm, E* = 1
Thickness Ratio = tr • E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

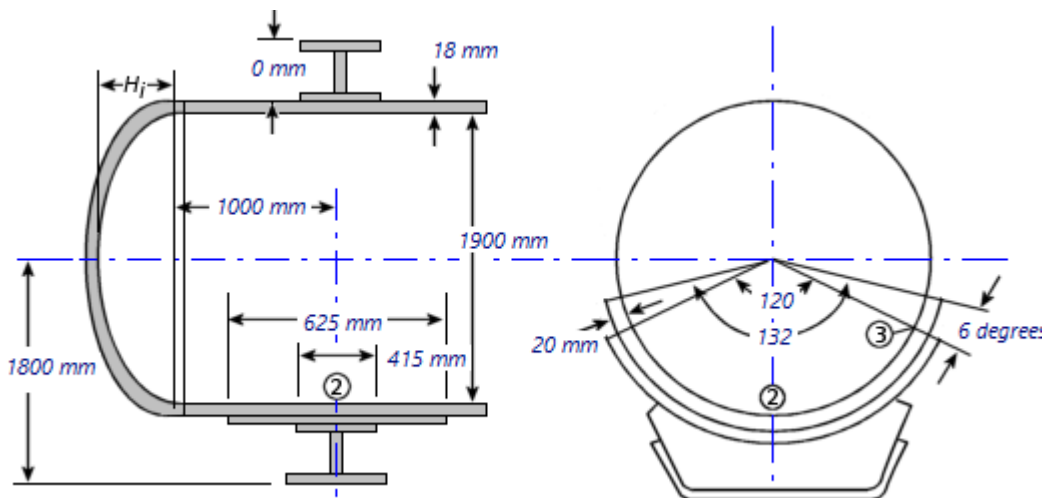
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	959.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	858.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	415.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		49766.43	kgf
Pressure used in Analysis	P	34.156	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		795.19	2386.01
Long. Stress at Bottom of Midspan		1024.58	2386.01
Long. Stress at Top of Saddles		966.39	2386.01
Long. Stress at Bottom of Saddles		878.57	2386.01
Tangential Shear in Shell		258.84	1413.05
Circ. Stress at Horn of Saddle		326.04	3579.01
Circ. Compressive Stress in Shell		16.06	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1674/2 + 0) \cdot 1800/1711$$

$$= 2642.0 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{F}_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(278, 0) \cdot 1800/6580$$

$$= 76.0 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 47124 + \max(76, 2642, 0, 0)$$

$$= 49766.4 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1771 (0 + \pi/4(1.94 \cdot 1.2)^2)$$

$$= 8258.274 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	50787.01	kgf
Transverse Shear Load	837.25	kgf
Longitudinal Shear Load	277.90	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1900/2 + 18 + 20)\sin(120/2), 2 \cdot 959)$$

$$= 1711.266 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -49766 \cdot 100 [1 - (1 - 100/858 + (95.9^2 - 0^2)/(2 \cdot 100 \cdot 858))/(1 + (4 \cdot 0)/(3 \cdot 858))]$$

$$= -3133.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 49766 \cdot 858/4((1 + 2(95.9^2 - 0^2)/(858^2))/(1 + (4 \cdot 0)/(3 \cdot 858)) - 4 \cdot 100/858)$$

$$= 59649.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 34.2 \cdot 959/(2 \cdot 18) - 59650/(\pi \cdot 959^2 \cdot 18)$$

$$= 795.19 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 34.2 \cdot 959/(2 \cdot 18) + 59650/(\pi \cdot 959^2 \cdot 18)$$

$$= 1024.58 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 34.2 \cdot 959/(2 \cdot 18) - 3133/(0.11 \cdot \pi \cdot 959^2 \cdot 18)$$

$$= 966.39 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 34.2 \cdot 959 / (2 \cdot 18) + 3133 / (0.19 \cdot \pi \cdot 959^2 \cdot 18)$$

$$= 878.57 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 49766 (858 - 2 \cdot 100) / (858 + (4 \cdot 0 / 3))$$

$$= 38165.9 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 38166 / (959 \cdot 18)$$

$$= 258.84 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(959 \cdot 18)}$$

$$= 102.480 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(415 + 1.56 \cdot \sqrt{(959 \cdot 18)}, 2 \cdot 100)$$

$$= 619.96 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 49766 \cdot 0.1 / (620(18 + 1 \cdot 20))$$

$$= -16.06 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -49766 / (4(18 + 1 \cdot 20)620) -$$

$$3 \cdot 0.053 \cdot 49766 / (2(18 + 1 \cdot 20)^2)$$

$$= -326.04 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	320.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	850.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(950 \cdot 18)} + 625)18^3) / 12$$

$$= 40.289 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 40.3 + 149(283 - 9)^2$$

$$= 111812.641 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	829.0	18.0	9.0	149.2	134297.4	0.112E+06
Wearplate	625.0	20.0	28.0	125.0	350000.0	0.811E+05
Web	20.0	784.0	430.0	156.8	6742398.0	0.114E+06
BasePlate	320.0	28.0	836.0	89.6	7490559.0	0.274E+06
Totals	...	...	...	520.6	14717254.0	0.582E+06



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 14717254/521 \\ &= 282.687 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 49766 \\ &= 10128.5479 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 27.2713 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1800 - 950 \cdot \sin(120/2) / 1.05 \\ &= 1014.356 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10273.9678 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 49.9323 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) \\ &= \sqrt{(3(49766 + 1021) \cdot 320) / (4 \cdot 1800 \cdot 1767)} \\ &= 19.574 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (950 + 18 + 20) \\ &= 1711.266 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1711 / (4 - 1) \\ &= 570.422 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 570 \cdot 320 / 2 \\ &= 912.675 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 50787 / (1800 \cdot 320) \\ &= 8.817 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 913 \cdot 8.82 \\ &= 8047.232 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 79 + 57 \\ &= 136.042 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 8047/136$$

$$= 59.153 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	415.0	...	83.0	...	0.119E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 415 - 20$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/136$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 415/2$$

$$= 207.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(11912/136)}$$

$$= 93.575 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1800 - \cos(120/2) (950 + 18 + 20) - 28$$

$$= 1278.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 1973951/2651)}$$

$$= 121.232$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1278/93.6$$

$$= 13.657$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 278 / (2 \cdot 1800) \cdot 570 \cdot 1278/2$$

$$= 28.137 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.7 < 121$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.7)^2 / (2 \cdot 121^2)) 2651 /$$

$$(5/3 + 3 \cdot (13.7) / (8 \cdot 121) - (13.7)^3 / (8 \cdot 121^3))$$

$$= 1542 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 59.2/1542 + (28.1 \cdot 208/119122272) / 1767$$

$$= 0.041$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	395.0	0.0	79.0	0.0	0.119E+05



Web	570.4	20.0	0.0	114.1	0.0	38.0
Totals	...	...	...	193.1	...	0.119E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/193$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 415/2$$

$$= 207.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1800 - \sqrt{(988 + 20)^2 - (570/2)^2} - 28$$

$$= 826.062 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(11949/193)}$$

$$= 78.667 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 826/78.7$$

$$= 10.501$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 278/(2 \cdot 1800)$$

$$= 0.077 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 570 \cdot 826$$

$$= 36.374 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 36.4/575852$$

$$= 6.317 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.5 < 121$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.5)^2 / (2 \cdot 121^2)) 2651 /$$

$$(5/3 + 3 \cdot (10.5) / (8 \cdot 121) - (10.5^3) / (8 \cdot 121^3))$$

$$= 1554 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 95.2/1554 + (36.4 \cdot 208/11949) / 1767$$

$$= 0.065$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	48145.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	76.0 kgf
Maximum Transverse Force	Ft	837.2 kgf
Maximum Longitudinal Force	Fl	277.9 kgf
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $278 / (6.46 \cdot 4)$
 = 10.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $837 / (6.46 \cdot 4)$
 = 32.4 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $278 / (1052 \cdot 4)$
 = 0.0660 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1800 \cdot 837 + 0$
 = 1507.05 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $1507 / 48145$
 = 31.30 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.066, 0]
 = 0.0660 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	959.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	858.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	415.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 50373.68 kgf



Pressure used in Analysis P 34.156 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	793.79	2386.01
Long. Stress at Bottom of Midspan	1025.98	2386.01
Long. Stress at Top of Saddles	967.08	2386.01
Long. Stress at Bottom of Saddles	878.18	2386.01
Tangential Shear in Shell	262.00	1413.05
Circ. Stress at Horn of Saddle	330.01	3579.01
Circ. Compressive Stress in Shell	16.26	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1674/2 + 0) \cdot 1800/1711$
 $= 2642.0\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(278, 0) \cdot 1800/6580$
 $= 76.0\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 47732 + \max(76, 2642, 0, 0)$
 $= 50373.7\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1771 (0 + \pi/4(1.94 \cdot 1.2)^2)$
 $= 8258.274\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	51394.27	kgf
Transverse Shear Load	837.25	kgf
Longitudinal Shear Load	277.90	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1900/2 + 18 + 20)\sin(120/2), 2 \cdot 959)$
 $= 1711.266\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -50374 \cdot 100 [1 - (1 - 100/858 + (95.9^2 - 0^2) / (2 \cdot 100 \cdot 858)) / (1 + (4 \cdot 0) / (3 \cdot 858))]$
 $= -3171.3\ kgf \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 50374 \cdot 858 / 4 ((1 + 2(95.9^2 - 0^2)/(858^2)) / (1 + (4 \cdot 0) / (3 \cdot 858)) - 4 \cdot 100/858)$
 $= 60377.7\ kgf \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 34.2 \cdot 959 / (2 \cdot 18) - 60378 / (\pi \cdot 959^2 \cdot 18)$$

$$= 793.79 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 34.2 \cdot 959 / (2 \cdot 18) + 60378 / (\pi \cdot 959^2 \cdot 18)$$

$$= 1025.98 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 34.2 \cdot 959 / (2 \cdot 18) - 3171 / (0.11 \cdot \pi \cdot 959^2 \cdot 18)$$

$$= 967.08 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 34.2 \cdot 959 / (2 \cdot 18) + 3171 / (0.19 \cdot \pi \cdot 959^2 \cdot 18)$$

$$= 878.18 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 50374 (858 - 2 \cdot 100) / (858 + (4 \cdot 0 / 3))$$

$$= 38631.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 38632 / (959 \cdot 18)$$

$$= 262.00 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(959 \cdot 18)}$$

$$= 102.480 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(415 + 1.56 \cdot \sqrt{(959 \cdot 18)}, 2 \cdot 100)$$

$$= 619.96 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot \tau_r))$$

$$= -0.76 \cdot 50374 \cdot 0.1 / (620(18 + 1 \cdot 20))$$

$$= -16.26 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [$\sigma_{7,r}$]:

$$= -Q / (4(t + \eta \cdot \tau_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot \tau_r)^2)$$

$$= -50374 / (4(18 + 1 \cdot 20) 620) -$$

$$3 \cdot 0.053 \cdot 50374 / (2(18 + 1 \cdot 20)^2)$$

$$= -330.01 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	320.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	850.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(950 \cdot 18)} + 625) 18^3) / 12$$

$$= 40.289 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 40.3 + 149 (283 - 9)^2$$

$$= 111812.641 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	829.0	18.0	9.0	149.2	134297.4	0.112E+06
Wearplate	625.0	20.0	28.0	125.0	350000.0	0.811E+05
Web	20.0	784.0	430.0	156.8	6742398.0	0.114E+06
BasePlate	320.0	28.0	836.0	89.6	7490559.0	0.274E+06
Totals	...	...	...	520.6	14717254.0	0.582E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14717254 / 521$$

$$= 282.687 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 50374$$

$$= 10252.1367 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 27.6041 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1800 - 950 \cdot \sin(120 / 2) / 1.05$$

$$= 1014.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10399.3320 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 50.5416 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(50374 + 1021) 320) / (4 \cdot 1800 \cdot 1767)}$$

$$= 19.691 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (950 + 18 + 20)$$

$$= 1711.266 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1711 / (4 - 1)$$

$$= 570.422 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 570 \cdot 320 / 2$$

$$= 912.675 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 51394 / (1800 \cdot 320)$$

$$= 8.923 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 913 \cdot 8.92$$

$$= 8143.452 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 79 + 57$$

$$= 136.042 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 8143 / 136$$

$$= 59.860 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	415.0	...	83.0	...	0.119E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 415 - 20$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 136$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 415 / 2$$

$$= 207.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{ (11912 / 136)}$$

$$= 93.575 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1800 - \cos(120 / 2) (950 + 18 + 20) - 28$$

$$= 1278.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2038903 / 2651)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1278 / 93.6$$

$$= 13.657$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 278 / (2 \cdot 1800) \cdot 570 \cdot 1278 / 2$$

= 28.137 kgf-m.

Compressive Allowable, $KL/r < Cc$ (13.7 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.7)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (13.7) / (8 \cdot 123) - (13.7^3) / (8 \cdot 123^3))$$

$$= 1543 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 59.9/1543 + (28.1 \cdot 208/119122272) / 1767$$

$$= 0.042$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	395.0	0.0	79.0	0.0	0.119E+05
Web	570.4	20.0	0.0	114.1	0.0	38.0
Totals	...	...	...	193.1	...	0.119E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/193$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 415/2$$

$$= 207.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1800 - \sqrt{ (988 + 20)^2 - (570/2)^2 } - 28$$

$$= 826.062 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 11949/193 }$$

$$= 78.667 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 826/78.7$$

$$= 10.501$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 278 / (2 \cdot 1800)$$

$$= 0.077 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 570 \cdot 826$$

$$= 36.374 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 36.4/575852$$

$$= 6.317 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (10.5 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.5)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (10.5) / (8 \cdot 123) - (10.5^3) / (8 \cdot 123^3))$$

$$= 1555 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$



$$= 96.4/1555 + (36.4 \cdot 208/11949)/1767$$

$$= 0.066$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	48752.3 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	76.0 kgf
Maximum Transverse Force	Ft	837.2 kgf
Maximum Longitudinal Force	F1	277.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 278 / (6.46 \cdot 4)$
 $= 10.8 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 837 / (6.46 \cdot 4)$
 $= 32.4 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 278 / (1052 \cdot 4)$
 $= 0.0660 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1800 \cdot 837 + 0$
 $= 1507.05 \text{ kgf-m}$

Eccentricity (e):
 $= Rmom / Q0$
 $= 1507/48752$
 $= 30.91 \text{ mm} < Bplen/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.066, 0]$
 $= 0.0660 \text{ cm}^2$

Review notes about nozzle loadings and supports in the Warnings report.

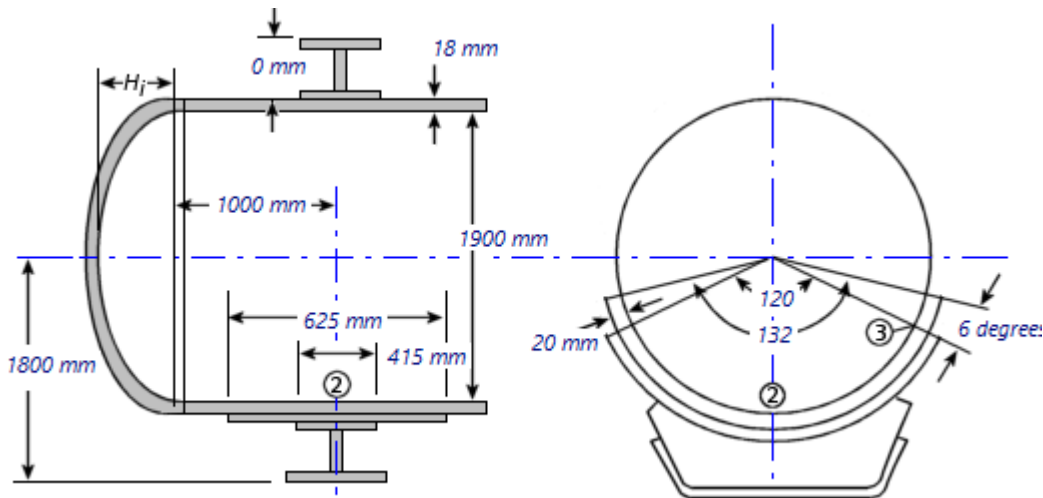
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	959.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	858.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	415.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	63925.75	kgf
Pressure used in Analysis	P	-1.055 kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-191.15	-2086.86	
Long. Stress at Top of Midspan	191.15	1407.13	
Long. Stress at Bottom of Midspan	129.81	1407.13	
Long. Stress at Top of Saddles	48.28	1407.13	
Long. Stress at Bottom of Saddles	-74.43	-2086.86	
Long. Stress at Bottom of Saddles	74.43	1407.13	
Tangential Shear in Shell	362.42	1125.71	
Circ. Stress at Horn of Saddle	452.02	1758.92	
Circ. Compressive Stress in Shell	21.78	1407.13	

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3 (5074/2 + 0) \cdot 1800/1711$
 $= 8006.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(842, 0) \cdot 1800/6580$
 $= 230.4 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(11236, 0) \cdot 1800/6580$
 $= 3073.7 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3 (11236/2 + 0) \cdot 1800/1711$
 $= 17728.1 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 46198 + \max(230, 8006, 3074, 17728)$
 $= 63925.8 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1771 (0 + \pi/4(1.94 \cdot 1.2)^2)$
 $= 8258.274 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	64946.33	kgf
Transverse Shear Load	5618.06	kgf
Longitudinal Shear Load	11236.12	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1900/2 + 18 + 20) \sin(120/2), 2 \cdot 960)$
 $= 1711.266 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$
 $= -63926 \cdot 100 [1 - (1 - 100/858 + (96^2 - 0^2)/$
 $(2 \cdot 100 \cdot 858)) / (1 + (4 \cdot 0)/(3 \cdot 858))]$
 $= -4019.1 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(\text{L}^2)) - 4a/L)$
 $= 63926 \cdot 858/4 ((1 + 2(96^2 - 0^2)/(858^2)) / (1 + (4 \cdot 0)/$
 $(3 \cdot 858)) - 4 \cdot 100/858)$
 $= 76626.5 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= -1.05 \cdot 960/(2 \cdot 16.5) - 76626/(\pi \cdot 960^2 \cdot 16.5)$
 $= -191.15 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 960 / (2 \cdot 16.5) + 76626 / (\pi \cdot 960^2 \cdot 16.5)$$

$$= 129.81 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 960 / (2 \cdot 16.5) - 4019 / (0.11 \cdot \pi \cdot 960^2 \cdot 16.5)$$

$$= 48.28 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 960 / (2 \cdot 16.5) + 4019 / (0.19 \cdot \pi \cdot 960^2 \cdot 16.5)$$

$$= -74.43 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 63926 (858 - 2 \cdot 100) / (858 + (4 \cdot 0 / 3))$$

$$= 49024.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 49025 / (960 \cdot 16.5)$$

$$= 362.42 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 16.5)}$$

$$= 98.156 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(415 + 1.56 \cdot \sqrt{(960 \cdot 16.5)}, 2 \cdot 100)$$

$$= 611.31 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407 / 1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 63926 \cdot 0.1 / (611(16.5 + 1 \cdot 20))$$

$$= -21.78 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -63926 / (4(16.5 + 1 \cdot 20)611) -$$

$$3 \cdot 0.053 \cdot 63926 / (2(16.5 + 1 \cdot 20)^2)$$

$$= -452.02 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	320.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	850.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(952 \cdot 16.5)} + 625)^3) / 12$$

$$= 30.714 \text{ cm}^4$$



Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 30.7 + 135 (289 - 8.25)²
 = 106631.727 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	820.5	16.5	8.3	135.4	111685.9	0.107E+06
Wearplate	625.0	20.0	26.5	125.0	331250.0	0.861E+05
Web	20.0	784.0	428.5	156.8	6718880.0	0.111E+06
BasePlate	320.0	28.0	834.5	89.6	7477121.0	0.267E+06
Totals	...	...	...	506.8	14638937.0	0.570E+06

Distance to Centroid [C1]:
 = AY / A
 = 14638937/507
 = 288.864 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.2 • 63926
 = 13010.2754 kgf

Tension Stress, St = (Fh/As) = 35.0304 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1391.2390 kgf/cm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1800 - 952 • sin(120/2)/1.05
 = 1013.116 mm

Bending Moment, M = Fh * d = 13180.9307 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 66.7489 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1545.8213 kgf/cm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))
 = √(3(63926 + 1021)320/(4 • 1800 • 1546))
 = 23.668 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120/2)(950 + 18 + 20)
 = 1711.266 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1711/(4 - 1)
 = 570.422 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 570 • 320/2
 = 912.675 cm²



Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 64946 / (1800 \cdot 320)$
 $= 11.275 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 913 \cdot 11.3$
 $= 10290.784 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 79 + 57$
 $= 136.042 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 10291 / 136$
 $= 75.644 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	415.0	...	83.0	...	0.119E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 415 - 20$
 $= 395.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 136$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 415 / 2$
 $= 207.500 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{ (11912 / 136)}$
 $= 93.575 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1800 - \cos(120 / 2) (950 + 18 + 20) - 28$
 $= 1278.000 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{ (2 \cdot \pi^2 \cdot 1973951 / 2319)}$
 $= 129.631$

Slenderness ratio [KL/r]:
 $= KL / r$
 $= 1 \cdot 1278 / 93.6$
 $= 13.657$

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 11236 / (2 \cdot 1800) \cdot 570 \cdot 1278 / 2$
 $= 1137.658 \text{ kgf-m.}$

Compressive Allowable, $KL/r < C_c$ ($13.7 < 130$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$



$$= (1 - (13.7)^2 / (2 \cdot 130^2)) 2319 /$$

$$(5/3 + 3 \cdot (13.7) / (8 \cdot 130) - (13.7^3) / (8 \cdot 130^3))$$

$$= 1352 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 75.6 / 1352 + (1138 \cdot 208 / 119122272) / 1546$$

$$= 0.184$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	395.0	0.0	79.0	0.0	0.119E+05
Web	570.4	20.0	0.0	114.1	0.0	38.0
Totals	...	...	...	193.1	...	0.119E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 193$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 415 / 2$$

$$= 207.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 1800 - \sqrt{(988 + 20)^2 - (570/2)^2} - 28$$

$$= 826.062 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(11949 / 193)}$$

$$= 78.667 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 826 / 78.7$$

$$= 10.501$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 11236 / (2 \cdot 1800)$$

$$= 3.121 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.12 \cdot 570 \cdot 826$$

$$= 1470.698 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1471 / 575852$$

$$= 255.395 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.5 < 130$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r^3) / (8 \cdot C_c^3))$$

$$= (1 - (10.5)^2 / (2 \cdot 130^2)) 2319 /$$

$$(5/3 + 3 \cdot (10.5) / (8 \cdot 130) - (10.5^3) / (8 \cdot 130^3))$$

$$= 1362 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 122 / 1362 + (1471 \cdot 208 / 11949) / 1546$$

$$= 0.255$$



Input and Calculated Values:

Vessel Mean Radius	Rm	959.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	858.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	415.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		64533.00	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-192.68	-2086.86
Long. Stress at Top of Midspan	192.68	1407.13
Long. Stress at Bottom of Midspan	131.34	1407.13
Long. Stress at Top of Saddles	49.03	1407.13
Long. Stress at Bottom of Saddles	-74.85	-2086.86
Long. Stress at Bottom of Saddles	74.85	1407.13
Tangential Shear in Shell	365.87	1125.71
Circ. Stress at Horn of Saddle	456.32	1758.92
Circ. Compressive Stress in Shell	21.99	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (5074/2 + 0) \cdot 1800/1711$
 $= 8006.0\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(842, 0) \cdot 1800/6580$
 $= 230.4\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(11236, 0) \cdot 1800/6580$
 $= 3073.7\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (11236/2 + 0) \cdot 1800/1711$
 $= 17728.1\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 46805 + \max(230, 8006, 3074, 17728)$
 $= 64533.0\ kgf$



Longitudinal Wind Force [F_L]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1771 (0 + \pi/4 (1.94 \cdot 1.2)^2)$
 $= 8258.274 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	65553.58	kgf
Transverse Shear Load	5618.06	kgf
Longitudinal Shear Load	11236.12	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1900/2 + 18 + 20) \sin(120/2), 2 \cdot 960)$
 $= 1711.266 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -64533 \cdot 100 [1 - (1 - 100/858 + (96^2 - 0^2)/(2 \cdot 100 \cdot 858)) / (1 + (4 \cdot 0)/(3 \cdot 858))]$
 $= -4057.3 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 64533 \cdot 858/4 ((1 + 2(96^2 - 0^2)/(858^2)) / (1 + (4 \cdot 0)/(3 \cdot 858)) - 4 \cdot 100/858)$
 $= 77354.4 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 $= -1.05 \cdot 960/(2 \cdot 16.5) - 77354/(\pi \cdot 960^2 \cdot 16.5)$
 $= -192.68 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 $= -1.05 \cdot 960/(2 \cdot 16.5) + 77354/(\pi \cdot 960^2 \cdot 16.5)$
 $= 131.34 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$
 $= -1.05 \cdot 960/(2 \cdot 16.5) - 4057/(0.11 \cdot \pi \cdot 960^2 \cdot 16.5)$
 $= 49.03 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$
 $= -1.05 \cdot 960/(2 \cdot 16.5) + 4057/(0.19 \cdot \pi \cdot 960^2 \cdot 16.5)$
 $= -74.85 \text{ kgf/cm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot h_m/3))$
 $= 64533 (858 - 2 \cdot 100) / (858 + (4 \cdot 0/3))$
 $= 49490.3 \text{ kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K2 \cdot T / (R_m \cdot t)$
 $= 1.17 \cdot 49490 / (960 \cdot 16.5)$
 $= 365.87 \text{ kgf/cm}^2$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 16.5)}$$

$$= 98.156 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(415 + 1.56 \cdot \sqrt{(960 \cdot 16.5)}, 2 \cdot 100)$$

$$= 611.31 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 64533 \cdot 0.1 / (611(16.5 + 1 \cdot 20))$$

$$= -21.99 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -64533 / (4(16.5 + 1 \cdot 20)611) -$$

$$3 \cdot 0.053 \cdot 64533 / (2(16.5 + 1 \cdot 20)^2)$$

$$= -456.32 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	320.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	850.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(952 \cdot 16.5)} + 625)16.5^3) / 12$$

$$= 30.714 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 30.7 + 135(289 - 8.25)^2$$

$$= 106631.727 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	820.5	16.5	8.3	135.4	111685.9	0.107E+06
Wearplate	625.0	20.0	26.5	125.0	331250.0	0.861E+05
Web	20.0	784.0	428.5	156.8	6718880.0	0.111E+06
BasePlate	320.0	28.0	834.5	89.6	7477121.0	0.267E+06
Totals	...	...	...	506.8	14638937.0	0.570E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14638937/507$$

$$= 288.864 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$



Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 64533 \\ &= 13133.8633 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 35.3632 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1800 - 952 \cdot \sin(120/2) / 1.05 \\ &= 1013.116 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 13306.1396 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 67.3829 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\ &= \sqrt{(3(64533 + 1021) \cdot 320) / (4 \cdot 1800 \cdot 1546)} \\ &= 23.779 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (950 + 18 + 20) \\ &= 1711.266 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1711 / (4 - 1) \\ &= 570.422 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 570 \cdot 320 / 2 \\ &= 912.675 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 65554 / (1800 \cdot 320) \\ &= 11.381 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 913 \cdot 11.4 \\ &= 10387.004 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 79 + 57 \\ &= 136.042 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 10387 / 136 \\ &= 76.351 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction



	B	D	Y	A	AY	Io
Rib+Web	20.0	415.0	...	83.0	...	0.119E+05

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 415 - 20
 = 395.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/136
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 415/2
 = 207.500 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{11912/136}$
 = 93.575 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1800 - cos(120/2)(950 + 18 + 20) - 28
 = 1278.000 mm

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2038903/2319)}$
 = 131.746

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1278/93.6
 = 13.657

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 11236 / (2 • 1800) • 570 • 1278/2
 = 1137.658 kgf-m.

Compressive Allowable, KL/r < Cc (13.7 < 132) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (13.7)^2 / (2 \cdot 132^2)) 2319 / (5/3 + 3 \cdot (13.7) / (8 \cdot 132) - (13.7^3) / (8 \cdot 132^3))$
 = 1352 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I)/Sba
 = 76.4/1352 + (1138 • 208/119122272)/1546
 = 0.185

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	395.0	0.0	79.0	0.0	0.119E+05
Web	570.4	20.0	0.0	114.1	0.0	38.0
Totals	...	...	...	193.1	...	0.119E+05

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/193
 = 0.000 mm

Distance to Centroid [C1]:



$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 415 / 2 \\ &= 207.500 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Length of Inner Rib [L]:} \\ &= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk} \\ &= 1800 - \sqrt{(988 + 20)^2 - (570/2)^2} - 28 \\ &= 826.062 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Radius of Gyration [r]:} \\ &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(11949/193)} \\ &= 78.667 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Slenderness ratio [KL/r]:} \\ &= \text{KL/r} \\ &= 1 \cdot 826 / 78.7 \\ &= 10.501 \end{aligned}$$

$$\begin{aligned} &\text{Unit Force [Force,u]:} \\ &= \text{FI} / (2 \cdot \text{Baseplate Length}) \\ &= 11236 / (2 \cdot 1800) \\ &= 3.121 \text{ kgf/mm} \end{aligned}$$

$$\begin{aligned} &\text{Moment at base of inner Rib [Mbase,c]:} \\ &= \text{Unit Force} \cdot e \cdot L \\ &= 3.12 \cdot 570 \cdot 826 \\ &= 1470.698 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Bending Stress due to Transverse Force and Weight Load } [\sigma_B, \text{base,c}]: \\ &= \text{Bending Moment} / \text{Section Modulus} \\ &= 1471 / 575852 \\ &= 255.395 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Compressive Allowable, } \text{KL/r} < \text{Cc} \text{ (} 10.5 < 132 \text{) per AISC E2-1 [Sca]:} \\ &= (1 - (\text{KL/r})^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL/r}) / (8 \cdot \text{Cc}) - (\text{KL/r})^3 / (8 \cdot \text{Cc}^3)) \\ &= (1 - (10.5)^2 / (2 \cdot 132^2)) 2319 / \\ &\quad (5/3 + 3 \cdot (10.5) / (8 \cdot 132) - (10.5)^3 / (8 \cdot 132^3)) \\ &= 1362 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{AISC Unity Check of Inside Ribs (must be } \leq 1 \text{)} \\ &= \text{Sc/Sca} + (\text{Mbase,c} \cdot \text{C1/I}) / \text{Sba} \\ &= 123 / 1362 + (1471 \cdot 208 / 11949) / 1546 \\ &= 0.255 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2-Tube Inlet

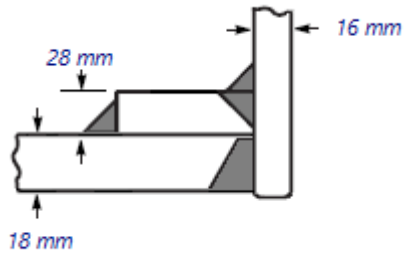
From: 30

Pressure for Reinforcement Calculations	P	10.387	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1900.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		127.42	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1494.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		350.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.4 \cdot 953 / (1407 \cdot 1 - 0.6 \cdot 10.4)$
= 7.0662 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.4 \cdot 384 / (1203 \cdot 1 - 0.6 \cdot 10.4)$
= 3.3324 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000 mm
Parallel to Vessel Wall, opening length	d	768.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203 / 1407)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203 / 1407)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407 / 1407)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.86, 1)$
= 0.855

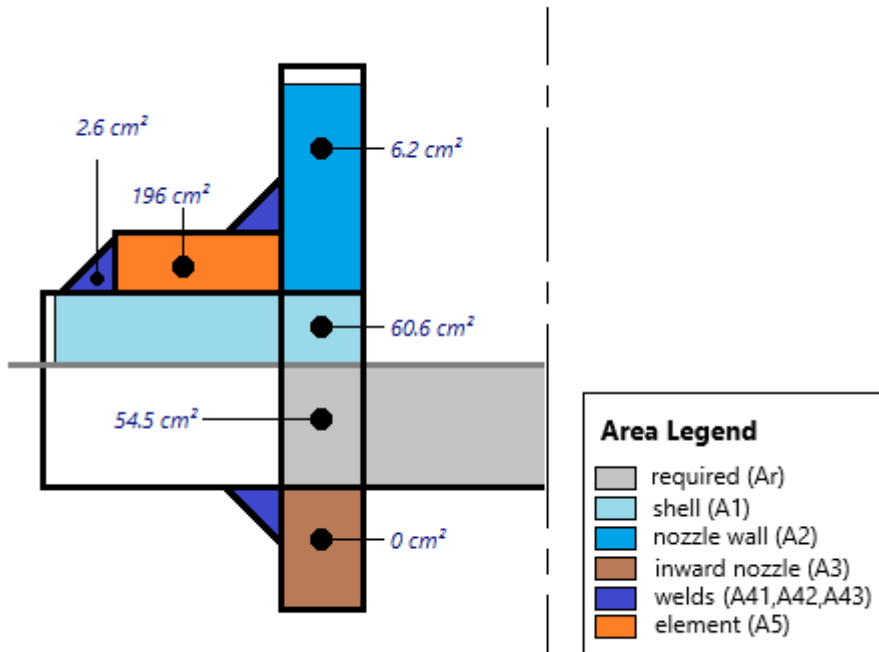


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	54.535	NA	NA
Area in Shell	A1	60.632	NA	NA
Area in Nozzle Wall	A2	6.200	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.618	NA	NA
Area in Element	A5	196.000	NA	NA
TOTAL AREA AVAILABLE	Atot	265.450	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (768 \cdot 7.07 \cdot 1 + 2 \cdot 13 \cdot 7.07 \cdot 1 \cdot (1 - 0.86))$
 $= 54.535 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 768 (1 \cdot 15 - 1 \cdot 7.07) - 2 \cdot 13$
 $(1 \cdot 15 - 1 \cdot 7.07) \cdot (1 - 0.86)$
 $= 60.632 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (13 - 3.33) \cdot 0.86$
 $= 6.200 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$
 $= (1.38) \cdot 0.86 + (0) \cdot 0.86 + 12^2 \cdot 1$



= 2.618 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
= (1494 - 794) * 28 * 1
= 196.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 6.3324 mm
Wall Thickness per UG16(b), tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 10.0662 mm
Wall Thickness tb1 = max(trb1, tr16b) = 10.0662 mm
Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45 tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[11.3, max(10.1, 5.5)]
= 10.0662 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(6.33, 10.1)
= 10.0662 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	331.6,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2676.5 kgf/cm ²	Passed
Occasional	:	150.9,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	122.8,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 16, tr = 3.33, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.26, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16, tr = 3.33, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.26, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18, tr = 7.07, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.47, Temp. Reduction = 36 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 3.33, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.26$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -47 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -47 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.39/19.99 = 0.520$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1152.0000 mm
 Parallel to Vessel Wall, opening length d 576.0000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		36.358	36.358
Area Available in Shell A1		30.167	30.167
Area Available in Nozzle Wall A2		5.373	6.200
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.231	1.178
Area Available in Pad A5		0.000	100.240
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		36.771	137.784

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 8.4840 = $0.7 \cdot W_o$ mm
 Pad Weld 7.5000 = $0.5 \cdot T_{minPad}$ 8.4840 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t-tr))S_v)$
 $= \max(0, (54.5 - 60.6 + 2 \cdot 13 \cdot 0.86 \cdot (1 \cdot 15 - 7.07))1407)$
 $= \max(0, -6097.77) \text{ kgf}$



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (6.2 + 196 + 2.62 - 0 \cdot 0.86) \cdot 1407 \\ &= 288205.69 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (6.2 + 0 + 1.23 + (3.33)) \cdot 1407 \\ &= 15149.17 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (6.2 + 0 + 2.62 + 196 + (3.33)) \cdot 1407 \\ &= 292898.16 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 794 \cdot 12 \cdot 0.49 \cdot 1203 \\ &= 88238. \text{ kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.14/2.0) \cdot 1494 \cdot 12 \cdot 0.49 \cdot 1407 \\ &= 194170. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 390) \cdot (16 - 3) \cdot 0.7 \cdot 1203 \\ &= 134323. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 794 \cdot 28 \cdot 0.74 \cdot 1407 \\ &= 363634. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 794 \cdot (14 - 0) \cdot 0.74 \cdot 1407 \\ &= 181817. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (194170 + 134323) = 328493 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (88238 + 363634 + 181817 + 0) = 633690 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (194170 + 181817 + 0) = 375988 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 328493 kgf, must exceed W = 0 kgf	or W1 = 288205 kgf
Path 2-2 = 633689 kgf, must exceed W = 0 kgf	or W2 = 15149 kgf
Path 3-3 = 375987 kgf, must exceed W = 0 kgf	or W3 = 292898 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18.5 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 86.9293 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 504.9293 mm

Warning:
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1900.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	16.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1494.000	mm
Thickness of Reinforcing Pad	28.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	10.39	kgf/cm ²
Radial Load	P	-2059.93	kgf
Circumferential Shear	Vc	3059.30	kgf
Longitudinal Shear	Vl	3059.30	kgf
Circumferential Moment	Mc	6880.65	kgf-m.
Longitudinal Moment	Ml	9174.19	kgf-m.
Torsional Moment	Mt	11467.69	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 794.0 + 2 \cdot 1.65 \cdot \sqrt{(974.5(18.0 - 3.0))}$$

$$= 1192.979 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1949.000	mm
Nozzle Outside Diameter	(d) =	794.000	mm
Vessel Thickness used	(T) =	43.000	mm
Nozzle Thickness used	(t) =	13.000	mm
T / t	=	3.308	
d / t	=	61.077	



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.743$$

Nr/P	=	0.031
Mr/P	=	0.048
M0/P	=	0.043
N0/P	=	0.190
MrD/Mc	=	0.124
NrDL/Mc	=	0.072
M0d/Mc	=	0.139
N0DL/Mc	=	0.327
MrD/Ml	=	0.058
NrDL/Ml	=	0.038
M0D/Ml	=	0.058
N0DL/Ml	=	0.393

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	21	21	21	21
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-245	-245	245	245
Outplane Bending (ML)	-216	216	216	-216
Normal Pressure Stress	225	235	225	235
Outplane Stress Summary	-186	198	736	257

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-23	-23	23	23
Inplane Bending (ML)	-219	219	219	-219
Inplane Pressure Stress	112	112	112	112
Inplane Stress Summary	-94	278	391	-111

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	26	26	26	26
Shear Stress Summary	32	32	21	21

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	196	290	737	371

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	21	21	21	21
Outplane Bending (P)	28	-28	28	-28



Outplane Membrane (Mc)	-153	-153	153	153
Outplane Bending (Mc)	-389	389	389	-389
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	112	112	112	112

Outplane Stress Summary	-380	341	705	-131
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	-33	-33	33	33
Inplane Bending (Mc)	-347	347	347	-347
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	225	235	225	235

Inplane Stress Summary	-120	520	642	-107
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	26	26	26	26

Shear Stress Summary	21	21	32	32
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	382	523	719	154
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	21	21	21	21
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-245	-245	245	245
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	362	374	362	374

Hoop Stress Summary	137	150	628	641
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	240	-240	240	-240
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-149	-149	149	149
Axial Bending (ML)	-1614	1614	1614	-1614
Axial Pressure Stress	180	180	180	180

Axial Stress Summary	-1336	1412	2192	-1518
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	18	18	-18	-18
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	89	89	89	89
Shear Stress Summary	107	107	70	70

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1490	1421	2195	2163

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	21	21	21	21
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-153	-153	153	153
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	362	374	362	374
Hoop Stress Summary	229	242	536	548

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	240	-240	240	-240
Axial Membrane (Mc)	-112	-112	112	112
Axial Bending (Mc)	-2699	2699	2699	-2699
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	180	180	180	180
Axial Stress Summary	-2384	2534	3239	-2640

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-18	-18	18	18
Shear due to Torsion	89	89	89	89
Shear Stress Summary	70	70	107	107

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2618	2536	3244	3196

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 794.0 + 2 \cdot 1.65 \cdot \sqrt{960.5(18.0 - 3.0)}$$

$$= 1190.103 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1921.000 mm
 Nozzle Outside Diameter (d) = 1494.000 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 363.000 mm
 T / t = 0.041
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 8.801$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.046
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.021
 MrD/Mc = 0.237
 NrDL/Mc = 0.092
 M0d/Mc = 0.066
 N0DL/Mc = 0.039
 MrD/Ml = 0.034
 NrDL/Ml = 0.035
 M0D/Ml = 0.011
 N0DL/Ml = 0.017

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	19	19	19	19
Outplane Bending (P)	101	-101	101	-101
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-45	-45	45	45
Outplane Bending (ML)	-184	184	184	-184
Normal Pressure Stress	654	665	654	665
Outplane Stress Summary	546	721	1005	443

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	42	42	42	42
Inplane Bending (P)	391	-391	391	-391
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-94	-94	94	94
Inplane Bending (ML)	-561	561	561	-561
Inplane Pressure Stress	327	327	327	327
Inplane Stress Summary	105	444	1416	-488

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21
Shear Stress Summary	30	30	13	13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Two * Max Shear Stress	548	724	1417	932
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	19	19	19	19
Outplane Bending (P)	101	-101	101	-101
Outplane Membrane (Mc)	-79	-79	79	79
Outplane Bending (Mc)	-805	805	805	-805
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	327	327	327	327

Outplane Stress Summary	-436	969	1333	-481
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	42	42	42	42
Inplane Bending (P)	391	-391	391	-391
Inplane Membrane (Mc)	-188	-188	188	188
Inplane Bending (Mc)	-2912	2912	2912	-2912
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	654	665	654	665

Inplane Stress Summary	-2012	3039	4190	-2408
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	21	21	21	21

Shear Stress Summary	13	13	30	30
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	2012	3039	4190	2408
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	225	235	225	235	225	235	225	235
Circ. Pl (SUS)	-224	-224	266	266	-30	-30	36	36
Circ. Q (SUS)	-188	188	244	-244	-315	315	379	-379
Long. Pm (SUS)	112	112	112	112	112	112	112	112
Long. Pl (SUS)	-20	-20	26	26	-132	-132	174	174
Long. Q (SUS)	-187	187	251	-251	-361	361	417	-417

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	26	26	26	26	26	26	26	26
Pm (SUS)	225.0	235.0	225.0	235.0	225.0	235.0	225.0	235.0
Pm+Pl (SUS)	92.3	92.3	491.1	501.1	215.2	225.2	287.0	287.5
Pm+Pl+Q (Total)	196.5	289.6	736.3	372.4	382.7	522.4	715.7	152.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	235.00	1407.13	Passed
Pm+Pl (SUS)	501.07	2110.70	Passed
Pm+Pl+Q (TOTAL)	736.27	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	654	665	654	665	654	665	654	665
Circ. Pl (SUS)	-26	-26	64	64	-146	-146	230	230
Circ. Q (SUS)	-83	83	285	-285	-2521	2521	3303	-3303
Long. Pm (SUS)	327	327	327	327	327	327	327	327
Long. Pl (SUS)	-52	-52	136	136	-60	-60	98	98
Long. Q (SUS)	-170	170	952	-952	-704	704	906	-906
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	654.0	665.0	654.0	665.0	654.0	665.0	654.0	665.0
Pm+Pl (SUS)	628.2	639.2	718.3	729.2	508.3	519.3	884.1	895.1
Pm+Pl+Q (Total)	546	725	1415	933	2013	3040	4187	2408

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	665.00	1407.13	Passed
Pm+Pl (SUS)	895.14	2110.70	Passed
Pm+Pl+Q (TOTAL)	4187.29	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	362	374	362	374	362	374	362	374
Circ. Pl (SUS)	-224	-224	266	266	-132	-132	174	174
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	180	180	180	180	180	180	180	180
Long. Pl (SUS)	-143	-143	155	155	-106	-106	118	118
Long. Q (SUS)	-1374	1374	1854	-1854	-2459	2459	2939	-2939
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	18	18	-18	-18	-18	-18	18	18
Shear Q (SUS)	89	89	89	89	89	89	89	89
Pm (SUS)	362.0	374.0	362.0	374.0	362.0	374.0	362.0	374.0
Pm+Pl (SUS)	141.1	152.8	629.1	641.1	232.0	243.9	537.4	549.3
Pm+Pl+Q (Total)	1490	1420	2192	2163	2618	2535	3241	3196

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	374.00	1203.20	Passed
Pm+Pl (SUS)	641.06	1804.80	Passed
Pm+Pl+Q (TOTAL)	3241.23	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	180	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1900.00	mm
Design Length of Section	L	8579.9980	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1033.42	cm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		14.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm

Flange Material		SA-105	
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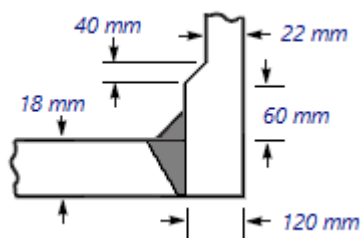
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 14.000 in.
Actual Thickness Used in Calculation 0.866 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.5 \cdot 952 / (1407 \cdot 1 - 0.6 \cdot 23.5)$
= 16.0198 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.5 \cdot 179 / (1203 \cdot 1 - 0.6 \cdot 23.5)$
= 3.5365 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.1422 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 98/0.5773
= 169.7410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

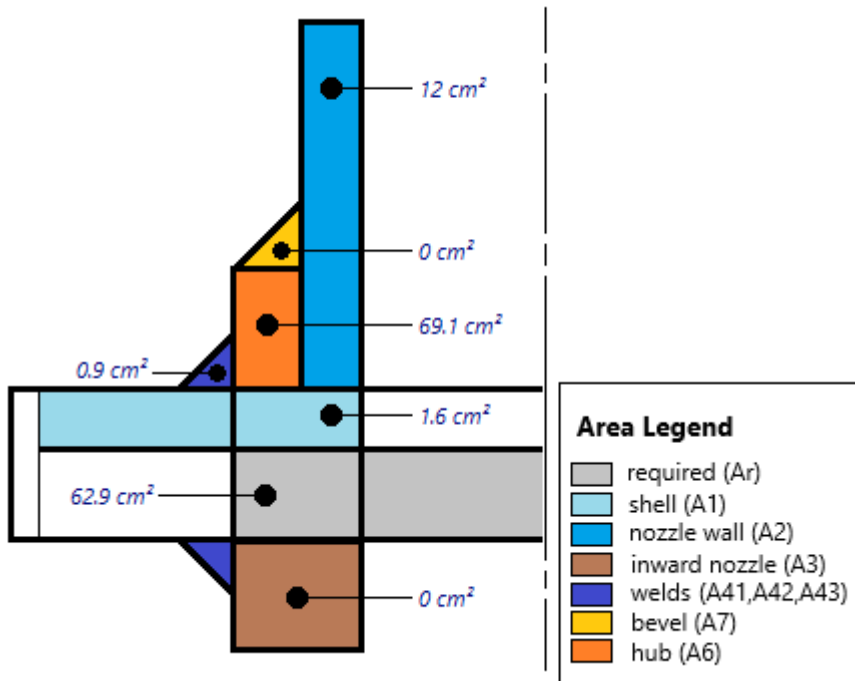
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	717.2000	mm
Parallel to Vessel Wall, opening length	d	358.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1203/1407)
= 0.855

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1203/1407)
= 0.855

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.86, 1)
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	62.949	22.896	NA
Area in Shell	A1	1.557	15.716	NA
Area in Nozzle Wall	A2	11.967	13.656	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	69.133	69.133	NA
TOTAL AREA AVAILABLE	Atot	83.512	99.359	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (359 \cdot 16 \cdot 1 + 2 \cdot 118 \cdot 16 \cdot 1 \cdot (1 - 0.86))$
 $= 62.949 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 359(1 \cdot 16.5 - 1 \cdot 16) - 2 \cdot 118$
 $(1 \cdot 16.5 - 1 \cdot 16) \cdot (1 - 0.86)$
 $= 1.557 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 41.2)(20.5 - 3.54)0.86$
 $= 11.967 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(41.2, 300, 60)) \cdot (120 - 22) \cdot 0.86$$

$$= 69.133 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0365 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.5198 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.5198 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.2134 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.83, \max(17.5, 3)]$$

$$= 9.8312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.04, 9.83)$$

$$= 9.8312 \text{ mm}$$

Available Nozzle Neck Thickness = 22.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 206.5,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2801.5 kgf/cm ²	Passed
Occasional	: 97.0,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 77.8,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 22, tr = 3.54, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.17, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 23.45/52.01 = 0.451

Weld Size Calculations, Description: N1-Shell Inlet

Intermediate Calc. for nozzle/shell welds Tmin 16.5000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (62.9 - 1.56 + 2 \cdot 118 \cdot 0.86 \cdot (1 \cdot 16.5 - 16)) \cdot 1407)$
 $= 87756.19 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (81.1 + 0.86 - 0 \cdot 0.86) \cdot 1407$
 $= 115320.62 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (81.1 + 0 + 0.86 + (33.4)) \cdot 1407$
 $= 162371.77 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (81.1 + 0 + 0.86 + 0 + (33.4)) \cdot 1407$
 $= 162371.77 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 596 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 55158. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 239) \cdot (120 - 1.5) \cdot 0.7 \cdot 1203$
 $= 747969. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 596 \cdot (18 - 1.5) \cdot 0.74 \cdot 1407$
 $= 160740. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (55158 + 747969) = 803127 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (55158 + 0 + 160740 + 0) = 215898 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (55158 + 160740 + 0) = 215898 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 803126 kgf, must exceed W = 87756 kgf	or W1 = 115320 kgf
Path 2-2 = 215898 kgf, must exceed W = 87756 kgf	or W2 = 162371 kgf
Path 3-3 = 215898 kgf, must exceed W = 87756 kgf	or W3 = 162371 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 47.8829 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 365.8829 mm

Input Echo, WRC297 Item 1, Description: N1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1900.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	355.600	mm
Nozzle Thickness	120.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	23.45	kgf/cm ²
Radial Load	P	-958.58	kgf
Circumferential Shear	Vc	1427.68	kgf
Longitudinal Shear	Vl	1427.68	kgf
Circumferential Moment	Mc	1498.45	kgf-m.
Longitudinal Moment	Ml	1997.94	kgf-m.
Torsional Moment	Mt	2497.41	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.963) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 595.6 + 2 \cdot 1.65 \cdot \sqrt{(959.75(18.0 - 1.5))}$$

$$= 1010.874 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1919.500 mm
Nozzle Outside Diameter	(d) =	595.600 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	118.500 mm
T / t	=	0.139
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.347$$

Nr/P	=	0.057
Mr/P	=	0.098
M0/P	=	0.029
N0/P	=	0.046
MrD/Mc	=	0.262
NrDL/Mc	=	0.101
M0d/Mc	=	0.079
N0DL/Mc	=	0.071
MrD/Ml	=	0.074
NrDL/Ml	=	0.067
M0D/Ml	=	0.022
N0DL/Ml	=	0.055

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	60	-60	60	-60
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-67	-67	67	67
Outplane Bending (ML)	-160	160	160	-160
Normal Pressure Stress	1340	1364	1340	1364
Outplane Stress Summary	1188	1413	1646	1227

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	20	20	20	20
Inplane Bending (P)	205	-205	205	-205
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-82	-82	82	82
Inplane Bending (ML)	-545	545	545	-545
Inplane Pressure Stress	670	670	670	670
Inplane Stress Summary	268	946	1524	22

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	27	27	27	27
Shear Stress Summary	36	36	17	17

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1189	1416	1648	1227

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	60	-60	60	-60



Outplane Membrane (Mc)	-65	-65	65	65
Outplane Bending (Mc)	-435	435	435	-435
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	670	670	670	670

Outplane Stress Summary	245	996	1248	256
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	20	20	20	20
Inplane Bending (P)	205	-205	205	-205
Inplane Membrane (Mc)	-93	-93	93	93
Inplane Bending (Mc)	-1453	1453	1453	-1453
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1340	1364	1340	1364

Inplane Stress Summary	20	2538	3113	-181
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	27	27	27	27

Shear Stress Summary	17	17	36	36
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	247	2538	3114	444
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	16	16	16	16
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-67	-67	67	67
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	60	31	60

Hoop Stress Summary	-19	8	116	144
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	2	8	35	18
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	24	13	116	144

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	16	16	16	16
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-65	-65	65	65
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	60	31	60
Hoop Stress Summary	-17	10	114	142

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-22	22	22	-22
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	-11	28	50	0

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	4	4

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	18	28	114	143

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	1340	1364	1340	1364	1340	1364	1340	1364
Circ. Pl (SUS)	-51	-51	83	83	-73	-73	113	113
Circ. Q (SUS)	-100	100	220	-220	-1248	1248	1658	-1658
Long. Pm (SUS)	670	670	670	670	670	670	670	670
Long. Pl (SUS)	-62	-62	102	102	-49	-49	81	81
Long. Q (SUS)	-340	340	750	-750	-375	375	495	-495
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	27	27	27	27	27	27	27	27
Pm (SUS)	1340	1364	1340	1364	1340	1364	1340	1364
Pm+Pl (SUS)	1289	1313	1423	1447	1267	1291	1453	1477
Pm+Pl+Q (Total)	1190	1415	1645	1227	247	2539	3111	442

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1364.00	1407.13	Passed
Pm+Pl (SUS)	1477.11	2110.70	Passed
Pm+Pl+Q (TOTAL)	3111.69	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	60	31	60	31	60	31	60
Circ. Pl (SUS)	-51	-51	83	83	-49	-49	81	81
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-11	-11	11	11	-8	-8	8	8
Long. Q (SUS)	-3	3	7	-7	-20	20	24	-24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	31.0	60.0	31.0	60.0	31.0	60.0	31.0	60.0
Pm+Pl (SUS)	24.1	9.2	114.0	143.0	25.1	11.2	112.0	141.0
Pm+Pl+Q (Total)	22.5	12.1	114.0	143.0	18.7	27.2	112.2	142.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	60.00	1203.20	Passed
Pm+Pl (SUS)	143.01	1804.80	Passed
Pm+Pl+Q (TOTAL)	143.03	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: N3-Shell Liquid Outlet

From: 40

Pressure for Reinforcement Calculations	P	23.568	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	180	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1900.00	mm
Design Length of Section	L	8579.9980	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 276.42 cm

User Entered Minimum Design Metal Temperature -33.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material SA-105

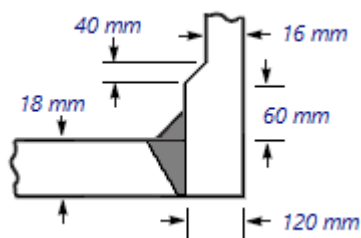
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class 300
 Flange Grade GR 1.1

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N3-Shell Liquid Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.630 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.6 \cdot 952 / (1407 \cdot 1 - 0.6 \cdot 23.6)$
= 16.0984 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.6 \cdot 103 / (1203 \cdot 1 - 0.6 \cdot 23.6)$
= 2.0435 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8410 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $88.6 / 0.5773$
= 153.4598 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

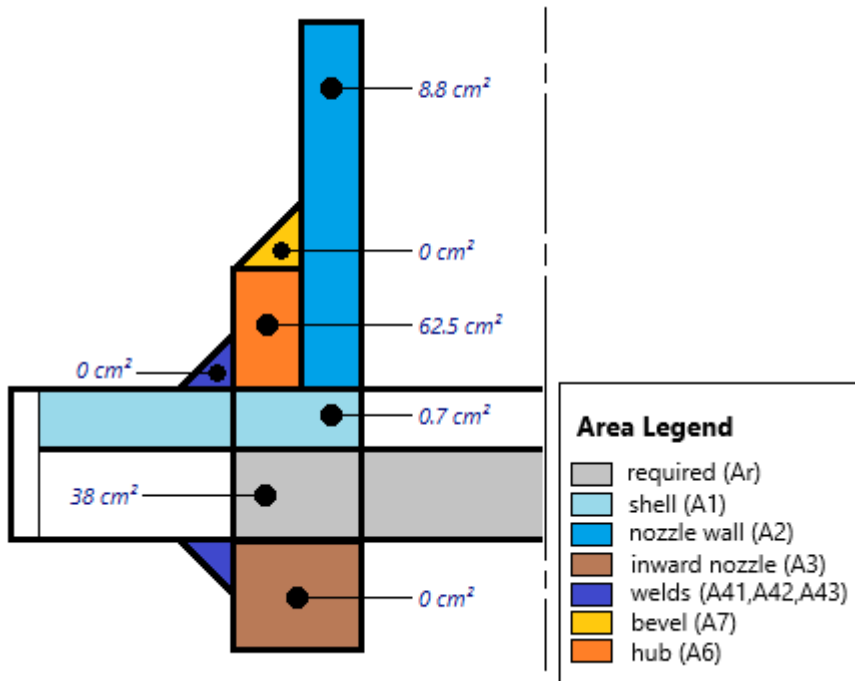
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	412.4000	mm
Parallel to Vessel Wall, opening length	d	206.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203 / 1407)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203 / 1407)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.86, 1)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	38.006	13.756	NA
Area in Shell	A1	0.708	8.546	NA
Area in Nozzle Wall	A2	8.787	9.636	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	62.502	62.502	NA
TOTAL AREA AVAILABLE	Atot	71.997	80.683	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (206 \cdot 16.1 \cdot 1 + 2 \cdot 103 \cdot 16.1 \cdot 1 \cdot (1 - 0.86))$$

$$= 38.006 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 206(1 \cdot 16.5 - 1 \cdot 16.1) - 2 \cdot 103$$

$$(1 \cdot 16.5 - 1 \cdot 16.1) \cdot (1 - 0.86)$$

$$= 0.708 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 41.2)(14.5 - 2.04)0.86$$

$$= 8.787 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can/0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10^2 - 1) \cdot 0.86 + (0^2 - 0) \cdot 0.86$$

$$= 0.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(41.2, 300, 60)) \cdot (105 - 16) \cdot 0.86 \\ &= 62.502 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5435 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.5984 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.5984 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.2134 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.6, \max(17.6, 3)] \\ &= 9.6026 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.54, 9.6) \\ &= 9.6026 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 228.8,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2779.2 kgf/cm ²	Passed
Occasional	: 78.3,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 108.8,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 16, tr = 2.04, c = 1.5 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.14, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.1, c = 1.5 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 23.57/52.01 = 0.453

Weld Size Calculations, Description: N3-Shell Liquid

Intermediate Calc. for nozzle/shell welds Tmin 16.5000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (38 - 0.71 + 2 \cdot 103 \cdot 0.86 \cdot (1 \cdot 16.5 - 16.1)) \cdot 1407)$
 $= 53479.03 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (71.3 + 0.73983\text{E-}06 - 0 \cdot 0.86) \cdot 1407$
 $= 100312.73 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (71.3 + 0 + 0.73983\text{E-}06 + (29.1)) \cdot 1407$
 $= 141249.22 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (71.3 + 0 + 0.73983\text{E-}06 + 0 + (29.1)) \cdot 1407$
 $= 141249.22 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 412 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 38192. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 155) \cdot (105 - 1.5) \cdot 0.7 \cdot 1203$
 $= 421885. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 412 \cdot (18 - 1.5) \cdot 0.74 \cdot 1407$
 $= 111298. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (38192 + 421885) = 460077 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (38192 + 0 + 111298 + 0) = 149490 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (38192 + 111298 + 0) = 149490 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 460077 kgf, must exceed W = 53479 kgf	or W1 = 100312 kgf
Path 2-2 = 149490 kgf, must exceed W = 53479 kgf	or W2 = 141249 kgf
Path 3-3 = 149490 kgf, must exceed W = 53479 kgf	or W3 = 141249 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 22.6481 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.6481 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Liqui:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1900.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	203.200	mm
Nozzle Thickness	120.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	23.57	kgf/cm ²
Radial Load	P	-550.67	kgf
Circumferential Shear	Vc	815.81	kgf
Longitudinal Shear	Vl	815.81	kgf
Circumferential Moment	Mc	489.29	kgf-m.
Longitudinal Moment	Ml	652.39	kgf-m.
Torsional Moment	Mt	815.48	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (3.693) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 443.2 + 2 \cdot 1.65 \cdot \sqrt{(959.75(18.0 - 1.5))}$$

$$= 858.474 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1919.500 mm
Nozzle Outside Diameter	(d) =	443.200 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	118.500 mm
T / t	=	0.139
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.490$$

Nr/P	=	0.069
Mr/P	=	0.124
M0/P	=	0.037
N0/P	=	0.055
MrD/Mc	=	0.280
NrDL/Mc	=	0.104
M0d/Mc	=	0.084
N0DL/Mc	=	0.071
MrD/Ml	=	0.094
NrDL/Ml	=	0.090
M0D/Ml	=	0.028
N0DL/Ml	=	0.074

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	44	-44	44	-44
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-39	-39	39	39
Outplane Bending (ML)	-90	90	90	-90
Normal Pressure Stress	1347	1370	1347	1370
Outplane Stress Summary	1273	1388	1533	1287

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	150	-150	150	-150
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-48	-48	48	48
Inplane Bending (ML)	-305	305	305	-305
Inplane Pressure Stress	673	673	673	673
Inplane Stress Summary	483	793	1192	280

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	15	15	15	15
Shear Stress Summary	23	23	8	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1273	1388	1533	1287

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	44	-44	44	-44



Outplane Membrane (Mc)	-28	-28	28	28
Outplane Bending (Mc)	-205	205	205	-205
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	673	673	673	673

Outplane Stress Summary	495	817	963	463
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	150	-150	150	-150
Inplane Membrane (Mc)	-42	-42	42	42
Inplane Bending (Mc)	-680	680	680	-680
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1347	1370	1347	1370

Inplane Stress Summary	789	1872	2234	596
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	15	15	15	15

Shear Stress Summary	8	8	23	23
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	789	1872	2234	599
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-39	-39	39	39
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	43	15	43

Hoop Stress Summary	-12	15	66	94
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	0	1	21	11
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	13	15	66	94

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-28	-28	28	28
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	43	15	43
Hoop Stress Summary	-2	26	55	83

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-10	10	10	-10
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-6	10	26	1

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	26	55	83

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	1347	1370	1347	1370	1347	1370	1347	1370
Circ. Pl (SUS)	-28	-28	50	50	-29	-29	55	55
Circ. Q (SUS)	-46	46	134	-134	-530	530	830	-830
Long. Pm (SUS)	673	673	673	673	673	673	673	673
Long. Pl (SUS)	-35	-35	61	61	-17	-17	39	39
Long. Q (SUS)	-155	155	455	-455	-161	161	249	-249
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	15	15	15	15	15	15	15	15
Pm (SUS)	1347	1370	1347	1370	1347	1370	1347	1370
Pm+Pl (SUS)	1319	1342	1397	1420	1318	1341	1402	1425
Pm+Pl+Q (Total)	1273	1388	1531	1286	788	1871	2232	598

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1370.00	1407.13	Passed
Pm+Pl (SUS)	1425.07	2110.70	Passed
Pm+Pl+Q (TOTAL)	2232.38	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	43	15	43	15	43	15	43
Circ. Pl (SUS)	-28	-28	50	50	-17	-17	39	39
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-7	-7	7	7	-5	-5	5	5
Long. Q (SUS)	-1	1	5	-5	-8	8	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	15.0	43.0	15.0	43.0	15.0	43.0	15.0	43.0
Pm+Pl (SUS)	13.0	15.0	65.0	93.0	4.0	26.0	54.0	82.0
Pm+Pl+Q (Total)	13.3	15.3	65.1	93.0	6.8	26.2	54.1	82.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.00	1203.20	Passed
Pm+Pl (SUS)	93.00	1804.80	Passed
Pm+Pl+Q (TOTAL)	93.05	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: N4-Shell Vapour Outlet

From: 40

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	180	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1900.00	mm
Design Length of Section	L	8579.9980	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		276.42	cm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	

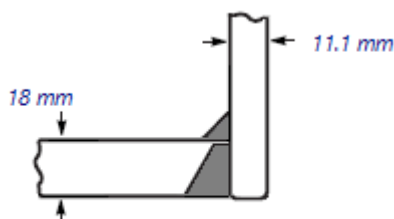
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4-Shell Vapour Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.503 in.
 Actual Thickness Used in Calculation 0.436 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.5 \cdot 952 / (1407 \cdot 1 - 0.6 \cdot 23.5)$
 $= 16.0198 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.5 \cdot 20.6 / (1203 \cdot 1 - 0.6 \cdot 23.5)$
 $= 0.4061 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3831 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	93.3250 mm
Parallel to Vessel Wall	Rn+tn+t	46.6625 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	23.9360 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-Shell Vapour.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9061 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.5198 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.5198 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.2134 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(17.5, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.91, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 362.0,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2646.0 kgf/cm ²	Passed
Occasional	: 20.5,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 298.1,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A



Govrn. thk, tg = 9.69, tr = 0.41, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.41, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $23.45 / 104.01 = 0.225$

Weld Size Calculations, Description: N4-Shell Vapour

Intermediate Calc. for nozzle/shell Welds Tmin 9.5744 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4789 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.4789 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Vapou:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1900.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	38.176	mm
Nozzle Thickness	11.0744	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	23.45	kgf/cm ²
Radial Load	P	-163.16	kgf
Circumferential Shear	Vc	254.94	kgf
Longitudinal Shear	Vl	254.94	kgf
Circumferential Moment	Mc	30.58	kgf-m.
Longitudinal Moment	Ml	50.97	kgf-m.
Torsional Moment	Mt	61.16	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.351) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.447) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(959.75(18.0 - 1.5))}$
 $= 475.599 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1919.500 mm
Nozzle Outside Diameter	(d) =	60.325 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	9.574 mm
T / t	=	1.723
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.339$$

Nr/P	=	0.229
Mr/P	=	0.261
M0/P	=	0.137
N0/P	=	0.178
MrD/Mc	=	0.248
NrDL/Mc	=	0.184
M0d/Mc	=	0.163
N0DL/Mc	=	-0.118
MrD/Ml	=	0.238
NrDL/Ml	=	0.183
M0D/Ml	=	0.155
N0DL/Ml	=	-0.067

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	20	20	-20	-20
Outplane Bending (ML)	-288	288	288	-288
Normal Pressure Stress	1340	1364	1340	1364
Outplane Stress Summary	1132	1635	1668	1016

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	93	-93	93	-93

Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-56	-56	56	56
Inplane Bending (ML)	-442	442	442	-442
Inplane Pressure Stress	670	670	670	670

Inplane Stress Summary	278	976	1277	204
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	16	16	-16	-16
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	64	64	64	64

Shear Stress Summary	81	81	48	48
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1140	1644	1674	1018
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	22	22	-22	-22
Outplane Bending (Mc)	-182	182	182	-182
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	670	670	670	670

Outplane Stress Summary	569	836	890	427
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	-34	-34	34	34
Inplane Bending (Mc)	-276	276	276	-276
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1340	1364	1340	1364

Inplane Stress Summary	1137	1526	1759	1041
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-16	-16	16	16
Torsional Shear (Mt)	64	64	64	64

Shear Stress Summary	48	48	81	81
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1141	1530	1766	1052

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	20	20	-20	-20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	49	77	49	77
Hoop Stress Summary	80	108	38	66

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	156	-156	156	-156
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-302	-302	302	302
Axial Bending (ML)	-808	808	808	-808
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	-918	385	1301	-626

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	28	28	-28	-28
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	111	111	111	111
Shear Stress Summary	139	139	83	83

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1037	443	1307	713

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	22	22	-22	-22
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	49	77	49	77
Hoop Stress Summary	81	109	37	65

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	10	10	10	10
Axial Bending (P)	156	-156	156	-156
Axial Membrane (Mc)	-181	-181	181	181
Axial Bending (Mc)	-516	516	516	-516
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24

Axial Stress Summary	-506	214	889	-456
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-28	-28	28	28
Shear due to Torsion	111	111	111	111
Shear Stress Summary	83	83	139	139

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	611	260	911	592

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1340	1364	1340	1364	1340	1364	1340	1364
Circ. Pl (SUS)	30	30	-10	-10	-21	-21	47	47
Circ. Q (SUS)	-239	239	337	-337	-183	183	369	-369
Long. Pm (SUS)	670	670	670	670	670	670	670	670
Long. Pl (SUS)	-43	-43	69	69	32	32	-12	-12
Long. Q (SUS)	-349	349	535	-535	-133	133	231	-231
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-16	-16	16	16
Shear Q (SUS)	64	64	64	64	64	64	64	64
Pm (SUS)	1340	1364	1340	1364	1340	1364	1340	1364
Pm+Pl (SUS)	1370	1394	1330	1354	1319	1343	1387	1411
Pm+Pl+Q (Total)	1138	1642	1672	1019	1140	1529	1763	1052

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1364.00	1407.13	Passed
Pm+Pl (SUS)	1411.34	2110.70	Passed
Pm+Pl+Q (TOTAL)	1763.32	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int.	Stress Values at
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Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	49	77	49	77	49	77	49	77
Circ. Pl (SUS)	30	30	-10	-10	32	32	-12	-12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-292	-292	312	312	-171	-171	191	191
Long. Q (SUS)	-652	652	964	-964	-360	360	672	-672
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	28	28	-28	-28	-28	-28	28	28
Shear Q (SUS)	111	111	111	111	111	111	111	111
Pm (SUS)	49.0	77.0	49.0	77.0	49.0	77.0	49.0	77.0
Pm+Pl (SUS)	351.5	379.2	338.6	338.9	234.8	262.1	219.3	220.1
Pm+Pl+Q (Total)	1036	441	1305	714	610	258	909	591

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	1203.20	Passed
Pm+Pl (SUS)	379.16	1804.80	Passed
Pm+Pl+Q (TOTAL)	1305.44	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1-Pressure Gauge

From: 40

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	180	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1900.00	mm
Design Length of Section	L	8579.9980	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		967.42	cm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	

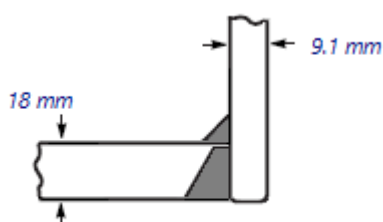
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1-Pressure Gauge

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 0.599 in.
Actual Thickness Used in Calculation 0.358 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.5 \cdot 952 / (1407 \cdot 1 - 0.6 \cdot 23.5)$
= 16.0198 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $9.11 \cdot (\exp([23.5 / (1203 \cdot 1)] - 1))$
= 0.1793 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2753 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	66.4010	mm
Parallel to Vessel Wall	Rn+tn+t	33.2005	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.9830	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1-Pressure Gau.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7753	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.5198	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.5198	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.2134	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.4464	mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[4.45, max(17.5, 3)]
= 4.4464 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.78, 4.45)
= 4.4464 mm

Available Nozzle Neck Thickness = $0.875 \cdot 9.093 = 7.957$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 7.96, tr = 0.18, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.028$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $23.45/52.01 = 0.451$

Weld Size Calculations, Description: K1-Pressure Gau

Intermediate Calc. for nozzle/shell Welds Tmin 7.5932 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.3152 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1468 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1468 mm

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Input, Nozzle Desc: N5-Tube Outlet

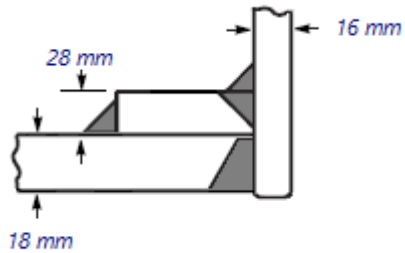
From: 50

Pressure for Reinforcement Calculations	P	10.197	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1900.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1197.42	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	450.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1494.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		350.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N5-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.2 \cdot 953 / (1407 \cdot 1 - 0.6 \cdot 10.2)$
= 6.9365 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.2 \cdot 384 / (1203 \cdot 1 - 0.6 \cdot 10.2)$
= 3.2711 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000 mm
Parallel to Vessel Wall, opening length	d	768.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203 / 1407)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203 / 1407)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407 / 1407)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.86, 1)$
= 0.855

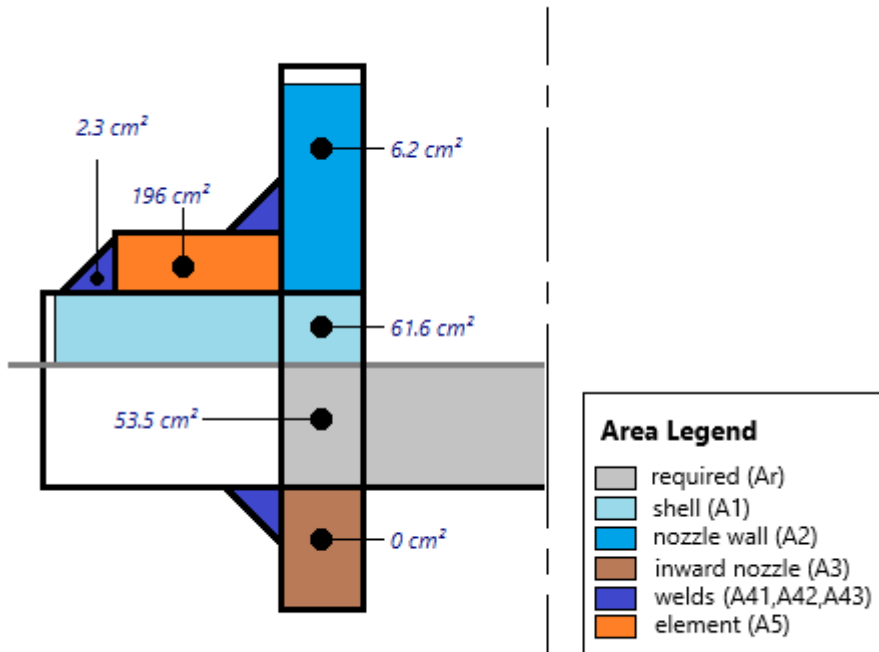


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	53.534	NA	NA
Area in Shell	A1	61.624	NA	NA
Area in Nozzle Wall	A2	6.239	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.293	NA	NA
Area in Element	A5	196.000	NA	NA
TOTAL AREA AVAILABLE	Atot	266.156	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (768 \cdot 6.94 \cdot 1 + 2 \cdot 13 \cdot 6.94 \cdot 1 \cdot (1 - 0.86))$
 $= 53.534 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 768(1 \cdot 15 - 1 \cdot 6.94) - 2 \cdot 13$
 $(1 \cdot 15 - 1 \cdot 6.94) \cdot (1 - 0.86)$
 $= 61.624 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (13 - 3.27) \cdot 0.86$
 $= 6.239 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$
 $= (1) \cdot 0.86 + (0) \cdot 0.86 + 12^2 \cdot 1$



= 2.293 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
= (1494 - 794) * 28 * 1
= 196.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 6.2711 mm
Wall Thickness per UG16(b), tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 9.9365 mm
Wall Thickness tb1 = max(trb1, tr16b) = 9.9365 mm
Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45 tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[11.3, max(9.94, 5.5)]
= 9.9365 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(6.27, 9.94)
= 9.9365 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	328.8,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2679.2 kgf/cm ²	Passed
Occasional	:	148.1,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	122.8,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 16, tr = 3.27, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16, tr = 3.27, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18, tr = 6.94, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.46, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 3.27, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E^* / (tg - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.20/19.99 = 0.510$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1152.0000 mm
 Parallel to Vessel Wall, opening length d 576.0000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		35.691	35.691
Area Available in Shell A1		30.660	30.660
Area Available in Nozzle Wall A2		5.407	6.239
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.855	0.853
Area Available in Pad A5		0.000	100.240
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		36.922	137.992

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N5-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm
 Pad Weld 7.5000 = $0.5 * T_{minPad}$ 8.4840 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (53.5 - 61.6 + 2 \cdot 13 \cdot 0.86 \cdot (1 \cdot 15 - 6.94)) 1407)$
 $= \max(0, -8861.70) \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$
$$= (6.24 + 196 + 2.29 - 0 \cdot 0.86) \cdot 1407$$
$$= 287803.72 \text{ kgf}$$

Weld Load [W2]:
$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$
$$= (6.24 + 0 + 0.86 + (3.33)) \cdot 1407$$
$$= 14675.02 \text{ kgf}$$

Weld Load [W3]:
$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$
$$= (6.24 + 0 + 2.29 + 196 + (3.33)) \cdot 1407$$
$$= 292496.19 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.14/2.0) \cdot 794 \cdot 10 \cdot 0.49 \cdot 1203$$
$$= 73532. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:
$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$
$$= (3.14/2.0) \cdot 1494 \cdot 12 \cdot 0.49 \cdot 1407$$
$$= 194170. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:
$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.14 \cdot 390) \cdot (16 - 3) \cdot 0.7 \cdot 1203$$
$$= 134323. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:
$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$
$$= (3.14/2) \cdot 794 \cdot 28 \cdot 0.74 \cdot 1407$$
$$= 363634. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:
$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$
$$= (3.14/2.0) \cdot 794 \cdot (14 - 0) \cdot 0.74 \cdot 1407$$
$$= 181817. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
$$= (SPEW + SNW) = (194170 + 134323) = 328493 \text{ kgf}$$

Failure Path [Path 2-2]:
$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (73532 + 363634 + 181817 + 0) = 618983 \text{ kgf}$$

Failure Path [Path 3-3]:
$$= (Spew + Tngw + Sinw)$$
$$= (194170 + 181817 + 0) = 375988 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 328493 kgf, must exceed W = 0 kgf	or W1 = 287803 kgf
Path 2-2 = 618983 kgf, must exceed W = 0 kgf	or W2 = 14675 kgf
Path 3-3 = 375987 kgf, must exceed W = 0 kgf	or W3 = 292496 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18.5 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 86.9293 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 554.9293 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 748.47 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm

Warning:
Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N5-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1900.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	16.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1494.000	mm
Thickness of Reinforcing Pad	28.0000	mm

Note:
*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	10.20	kgf/cm ²
Radial Load	P	-2059.93	kgf
Circumferential Shear	Vc	3059.30	kgf
Longitudinal Shear	Vl	3059.30	kgf
Circumferential Moment	Mc	6880.65	kgf-m.
Longitudinal Moment	Ml	9174.19	kgf-m.
Torsional Moment	Mt	11467.69	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 794.0 + 2 \cdot 1.65 \cdot \sqrt{(974.5(18.0 - 3.0))}$
 $= 1192.979 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1949.000	mm
Nozzle Outside Diameter	(d) =	794.000	mm
Vessel Thickness used	(T) =	43.000	mm



Nozzle Thickness used (t) = 13.000 mm
 T / t = 3.308
 d / t = 61.077

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.743$$

Nr/P = 0.031
 Mr/P = 0.048
 M0/P = 0.043
 N0/P = 0.190
 MrD/Mc = 0.124
 NrDL/Mc = 0.072
 M0d/Mc = 0.139
 N0DL/Mc = 0.327
 MrD/M1 = 0.058
 NrDL/M1 = 0.038
 M0D/M1 = 0.058
 N0DL/M1 = 0.393

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	21	21	21	21
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-245	-245	245	245
Outplane Bending (ML)	-216	216	216	-216
Normal Pressure Stress	220	231	220	231
Outplane Stress Summary	-190	194	732	252

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-23	-23	23	23
Inplane Bending (ML)	-219	219	219	-219
Inplane Pressure Stress	110	110	110	110
Inplane Stress Summary	-96	276	389	-113

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	26	26	26	26
Shear Stress Summary	32	32	21	21

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	200	288	733	369

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	21	21	21	21
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	-153	-153	153	153
Outplane Bending (Mc)	-389	389	389	-389
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	110	110	110	110
Outplane Stress Summary	-382	339	703	-133

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	-33	-33	33	33
Inplane Bending (Mc)	-347	347	347	-347
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	220	231	220	231
Inplane Stress Summary	-124	516	638	-111

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	26	26	26	26
Shear Stress Summary	21	21	32	32

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	384	518	716	157

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	21	21	21	21
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-245	-245	245	245
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	355	367	355	367
Hoop Stress Summary	131	143	621	634

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	240	-240	240	-240
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-149	-149	149	149



Axial Bending (ML)	-1614	1614	1614	-1614
Axial Pressure Stress	177	177	177	177

Axial Stress Summary	-1340	1408	2188	-1521
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	18	18	-18	-18
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	89	89	89	89
Shear Stress Summary	107	107	70	70

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1486	1417	2191	2160

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	21	21	21	21
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-153	-153	153	153
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	355	367	355	367
Hoop Stress Summary	223	235	529	542

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	240	-240	240	-240
Axial Membrane (Mc)	-112	-112	112	112
Axial Bending (Mc)	-2699	2699	2699	-2699
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	177	177	177	177
Axial Stress Summary	-2387	2531	3236	-2643

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-18	-18	18	18
Shear due to Torsion	89	89	89	89
Shear Stress Summary	70	70	107	107

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	2614	2533	3240	3193

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 794.0 + 2 \cdot 1.65 \cdot \sqrt{(960.5(18.0 - 3.0))}$$

$$= 1190.103 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1921.000 mm
 Nozzle Outside Diameter (d) = 1494.000 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 363.000 mm
 T / t = 0.041
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 8.801$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.046
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.021
 MrD/Mc = 0.237
 NrDL/Mc = 0.092
 M0d/Mc = 0.066
 N0DL/Mc = 0.039
 MrD/M1 = 0.034
 NrDL/M1 = 0.035
 M0D/M1 = 0.011
 N0DL/M1 = 0.017

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	19	19	19	19
Outplane Bending (P)	101	-101	101	-101
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-45	-45	45	45
Outplane Bending (ML)	-184	184	184	-184
Normal Pressure Stress	642	653	642	653
Outplane Stress Summary	534	709	993	430

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	42	42	42	42
Inplane Bending (P)	391	-391	391	-391
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-94	-94	94	94
Inplane Bending (ML)	-561	561	561	-561
Inplane Pressure Stress	321	321	321	321
Inplane Stress Summary	99	438	1410	-494

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21



Shear Stress Summary 30 30 13 13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	536	712	1411	926

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	19	19	19	19
Outplane Bending (P)	101	-101	101	-101
Outplane Membrane (Mc)	-79	-79	79	79
Outplane Bending (Mc)	-805	805	805	-805
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	321	321	321	321

Outplane Stress Summary -442 963 1327 -487

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	42	42	42	42
Inplane Bending (P)	391	-391	391	-391
Inplane Membrane (Mc)	-188	-188	188	188
Inplane Bending (Mc)	-2912	2912	2912	-2912
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	642	653	642	653

Inplane Stress Summary -2024 3027 4178 -2420

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	21	21	21	21

Shear Stress Summary 13 13 30 30

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2024	3027	4178	2421

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	220	231	220	231	220	231	220	231
Circ. Pl (SUS)	-224	-224	266	266	-30	-30	36	36
Circ. Q (SUS)	-188	188	244	-244	-315	315	379	-379

Long. Pm (SUS)	110	110	110	110	110	110	110	110
Long. Pl (SUS)	-20	-20	26	26	-132	-132	174	174
Long. Q (SUS)	-187	187	251	-251	-361	361	417	-417
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	26	26	26	26	26	26	26	26
Pm (SUS)	220.0	231.0	220.0	231.0	220.0	231.0	220.0	231.0
Pm+Pl (SUS)	94.5	90.3	486.1	497.1	212.2	223.2	284.9	285.4
Pm+Pl+Q (Total)	201.2	287.4	731.3	370.4	384.7	518.5	713.3	155.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	231.00	1407.13	Passed
Pm+Pl (SUS)	497.07	2110.70	Passed
Pm+Pl+Q (TOTAL)	731.28	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	642	653	642	653	642	653	642	653
Circ. Pl (SUS)	-26	-26	64	64	-146	-146	230	230
Circ. Q (SUS)	-83	83	285	-285	-2521	2521	3303	-3303
Long. Pm (SUS)	321	321	321	321	321	321	321	321
Long. Pl (SUS)	-52	-52	136	136	-60	-60	98	98
Long. Q (SUS)	-170	170	952	-952	-704	704	906	-906
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	642.0	653.0	642.0	653.0	642.0	653.0	642.0	653.0
Pm+Pl (SUS)	616.2	627.2	706.3	717.2	496.3	507.3	872.1	883.1
Pm+Pl+Q (Total)	534	713	1409	927	2025	3028	4175	2420

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	653.00	1407.13	Passed
Pm+Pl (SUS)	883.14	2110.70	Passed
Pm+Pl+Q (TOTAL)	4175.29	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	355	367	355	367	355	367	355	367
Circ. Pl (SUS)	-224	-224	266	266	-132	-132	174	174
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	177	177	177	177	177	177	177	177
Long. Pl (SUS)	-143	-143	155	155	-106	-106	118	118
Long. Q (SUS)	-1374	1374	1854	-1854	-2459	2459	2939	-2939
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	18	18	-18	-18	-18	-18	18	18
Shear Q (SUS)	89	89	89	89	89	89	89	89
Pm (SUS)	355.0	367.0	355.0	367.0	355.0	367.0	355.0	367.0
Pm+Pl (SUS)	134.2	145.9	622.1	634.1	225.1	237.0	530.4	542.3
Pm+Pl+Q (Total)	1486	1416	2189	2159	2614	2532	3238	3192

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	367.00	1203.20	Passed
Pm+Pl (SUS)	634.07	1804.80	Passed
Pm+Pl+Q (TOTAL)	3238.23	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	23.45	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	23.45	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05	kgf/cm ²
Shell Thickness	ts	18.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1900.000	mm
Shell Circumferential Joint Efficiency	Esw	1.000	
Shell Temperature for Internal Pressure	Ts	180.00	°C
Shell Material		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1407.13	kgf/cm ²
Shell Allowable Stress at Ambient		1407.13	kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	10.20	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	10.20	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	18.0000	mm
Channel Internal Corrosion Allowance	cac	3.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1900.000	mm
Channel Design Temperature	TEMPC	80.00	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1407.13	kgf/cm ²
Channel Allowable Stress at Ambient		1407.13	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	4760	
Tube Wall Thickness	et	2.0000	mm
Tube Outside Diameter	D	19.0000	mm
Total Straight Tube Length	Lt	8700.00	mm
Straight Tube Length (bet. inner tubsht faces) L		8580.00	mm
Design Temperature of the Tubes		180.00	°C
Tube Material		SA-334 6	
Tube Material UNS Number		K03006	
Is this a Welded Tube		No	
Tube Material Specification used	Smls. & wld. tube		
Tube Allowable Stress at Temperature		1203.20	kgf/cm ²
Tube Allowable Stress At Ambient		1203.20	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2135.19	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000	mm
Tube Layout Pattern		Triangular	

Fillet Weld Leg	af	2.0000	mm
Groove Weld Leg	ag	2.0000	mm
Tube-Tubesheet Joint Weld Type	Partial	Strength	
Method for Tube-Tubesheet Jt. Allowable		UW-20.3	
Is the Tube Inset		No	

Radius to Outermost Tube Hole Center	ro	918.900	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 60.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 1936.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

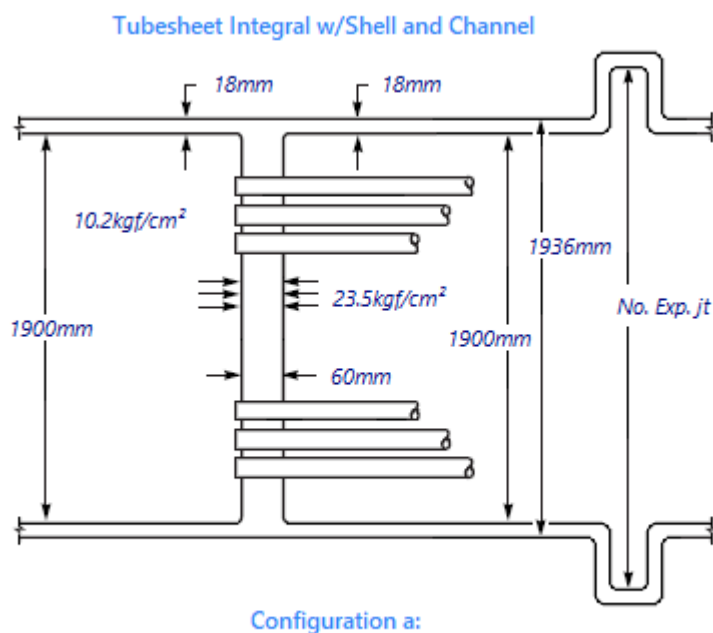
Unsupported Tube Span under consideration l 760.000 mm
 Tube End condition corresponding to Span (l) k 0.80

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Shell Side Min. Design Pressure Psd,min -1.0546 kgf/cm²

Mean Shell Metal Temp. along Shell len. Tsm 0.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 0.00 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%



Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20812E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20812E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (11.3 \cdot (19/2 - 0)) / (1203 \cdot 1 + 0.4 \cdot 11.3)$
 $= 0.0885 + 0.0000 = 0.0885 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	19.00	8580.00	9.50	50.0000	0.0121884	1218.00
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 176 \text{ kgf/cm}^2$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4787	19.00	8580.00	39.69	50.0000	0.0006983	698.17
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 698) / (3 \cdot 39.7) = 23.5 \text{ kgf/cm}^2$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4787 mm
Allowable Internal Pressure at Corroded thickness	276.60 kgf/cm ²
Required Internal Design Pressure	11.25 kgf/cm ²
Allowable External Pressure at Corroded thickness	176.37 kgf/cm ²
Required External Design Pressure	23.45 kgf/cm ²
Required Thickness due to Shell Side pressure	0.4787 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:



Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 0/60 \\ &= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(19 - 2 \cdot 2 (1973947/1973947) \cdot \\ &\quad (1203/1407) \cdot (0); 19 - 2 \cdot 2) \\ &= 19.0000 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 919 + 19 \\ &= 1856.800 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (25 - 19) / 25 \\ &= 0.240 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= D_o / 2 \\ &= 1857/2 \\ &= 928.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 25 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1857 \cdot 25) / (\pi \cdot 1857^2)) } \\ &= 25.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (25 - 19) / 25 \\ &= 0.2400 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= a_s / a_o \\ &= 950/928 \\ &= 1.0233 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= a_c / a_o \\ &= 950/928 \\ &= 1.0233 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\ &= 1 - 4760 ((19 - 2 \cdot 2) / (2 \cdot 928))^2 \\ &= 0.6894 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot a_o))^2 \\ &= 1 - 4760 (19 / (2 \cdot 928))^2 \\ &= 0.5016 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((8 - 0); 0) \\ &= 8.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s(D_s + t_s) E_s / L$$



$$= \pi \cdot 18 (1900 + 18) 1973947/8580$$

$$= 249527.6562 \text{ kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2 (19 - 2) 1973947/8580$$

$$= 245.7403 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 249528 / (4760 \cdot 246)$$

$$= 0.21332$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_j)$$

$$= 1 / (1 + 249528/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{(D_s + t_s)t_s}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1900 + 18)18}$$

$$= 0.0098 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0098 \cdot 1973947 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 20627920.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(1900 + 18) \cdot 20627920/60^3 \cdot (1 + 60 \cdot 0.0098 + (60 \cdot 0.0098)^2/2)$$

$$= 1933490.3750 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1900^2 / (4 \cdot 1973947 \cdot 18) \cdot (1 - 1900 \cdot 0.3 / (2(1900 + 18)))$$

$$= 0.21626\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1900 + 18)18}$$

$$= 0.0098 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0098 \cdot 2032634 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 21241204.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c/\cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(1900 + 18/\cos(0)) \cdot 21241204/60^3 \cdot (1 + 60 \cdot 0.0098 \cdot (1 + 1)/2 + (60 \cdot 0.0098)^2/2 \cdot 1)$$

$$= 1990974.5 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1900^2 / (4 \cdot 2032634 \cdot 18) \cdot (1 - 1900 \cdot 0.3 / (2(1900 + 18)))$$

$$= 0.21002\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 2.400000 ; \mu^* = 0.240000 ; E^* = 402743 \text{ kgf/cm}^2$$

$$E^*/E = 0.204029 ; \nu^* = 0.407183$$



Note: As h/p (2.400) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\frac{24(1 - \nu^2)Nt \cdot Et \cdot tt(dt - tt)ao^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.41^2)4760 \cdot 1973947 \cdot 2}{(19 - 2)928^2(402743 \cdot 8580 \cdot 60^3)} \right)^{1/4}$$

$$= 9.2705$$

Values from Table 4.18.3

$$Zd = 0.001846 ; Zv = 0.011692 ; Zm = 0.154073$$

$$Za = 0.561019E+04 ; Zw = 0.011692$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1936/1857$$

$$= 1.0427$$

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 402743 \cdot (1933490 + 1990974 + 1973947 \cdot \ln(1.04))$$

$$= 5.8980$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.41) 5.9$$

$$= 8.2995$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.02 - 1 - 8.3 \cdot 0.012) / (1 + 8.3 \cdot 0.15)$$

$$= -0.032375$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.0018 + -0.032 \cdot 0.012) / 2 \cdot 9.27^4$$

$$= 5.4204$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.012 + -0.032 \cdot 0.15) / 2 \cdot 9.27^4$$

$$= 24.7580$$

Compute parameter [U]:

$$= (Zw + (\psi_s - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.012 + (1.02 - 1)0.15)9.27^4 / (1 + 8.3 \cdot 0.15)$$

$$= 49.5160$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 20627920 \cdot 0.0098 \cdot 0.021626027(1 + 60 \cdot 0.0098)$$

$$= 7087.5430 \text{ mm}^2$$

Calculate parameter [ω_s]:

$$= ao^2(\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s$$

$$= 928^2(1.02^2 - 1) \cdot (1.02 - 1) / 4 - 7088$$

$$= -6851.5488 \text{ mm}^2$$

Parameter [ω_c]:



$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta_c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 21241204 [0.021001635 (1 + 0.0098 \cdot 1 \cdot 60) - 0.000 (1/60 + 1 \cdot 0.0098/2)]$$

$$= 7087.5435 \text{ mm}^2$$

Parameter $[\omega \cdot c]$:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1)/4 - (\rho_s - 1)/2] - \omega_c$$

$$= 928^2 [(1.02^2 + 1) \cdot (1.02 - 1)/4 - (1.02 - 1)/2] - 7088$$

$$= -6851.5503 \text{ mm}^2$$

Pressure $[P \cdot S]$:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure $[P \cdot C]$:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure $[P's]$:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(\rho_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 23.5 \{ 0.5 + 2(1 - 0.5) 0.3 + [2/0.21 \cdot (1900/1857)^2] 0.3 - [(1.02^2 - 1) / (1 \cdot 0.21)] - [(1 - 1) / (2 \cdot 1 \cdot 0.21)] [(0^2 - (1900)^2) / 1857^2] \}$$

$$= 82.6760 \text{ kgf/cm}^2$$

Pressure $[P't]$:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.69 + 2(1 - 0.69) 0.3 + 1/(1 \cdot 0.21)] 10.2$$

$$= 56.7332 \text{ kgf/cm}^2$$

Pressure $[P_y]$:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 4760 \cdot 246 \cdot 0 / (3.14 \cdot 928^2)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Pressure $[P_w]$:

$$= -\gamma_b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 49.5 \cdot 0 / (2 \cdot \pi \cdot 928^2)$$

$$= -0.0000 \text{ kgf/cm}^2$$

Calculate the pressure $[Prim]$:

$$= -(U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= -(49.5/928^2) \cdot (-6852 \cdot 23.5 - -6852 \cdot 10.2)$$

$$= 5.2179 \text{ kgf/cm}^2$$

Calculate the pressure $[P_w]$:

$$= (U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= (49.5/928^2) \cdot (7088 \cdot 0 - 7088 \cdot 0)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure $[Pe]$:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho_s - 1) Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.21 / (1 + 1 \cdot 0.21 (5.42 + (1.02 - 1) 24.8)) \cdot (82.7 - 56.7 + 0 + 0 + -0 + 5.22)$$

$$= 2.9165 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor $[Q2]$:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-6852 \cdot 23.5 - -6852 \cdot 10.2) - (7088 \cdot 0 - 7088 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 8.3 \cdot 0.15)$$

$$= -398.5905 \text{ kgf}$$

Calculate factor $[Q3]$:



$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ &= -0.032 + 2 \cdot -399 / (2.92 \cdot 928^2) \\ &= -0.06409 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.032043

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.032 / 0.24) \cdot (2 \cdot 928 / (60 - 8))^2 \cdot 2.92 \\ &= 744.7452 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1407 = 2111 \text{ kgf/cm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.024 / 0.24) \cdot (2 \cdot 928 / (33.3 - 8))^2 \cdot 2.57 \\ &= 2110.5862 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 33.3 + 0 + 0 \\ &= 33.2906 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\ &= (2 \cdot 0.24 \cdot 60 / 928) \cdot 1126 \\ &= 34.9207 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Tau limit

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

$$\begin{aligned} \text{Tubesheet thickness} &= 5.0111 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 445.8396 \text{ kgf/cm}^2 \end{aligned}$$

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$\begin{aligned} &= \max(HreqB, HreqS) \\ &= \max(33.3, 5.01) \\ &= 33.2906 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
F_{tmin} = -0.25, F_{tmax} = 4.05

First Extreme Tube Axial Stress from among all the tubes [σ_1]:

$$\begin{aligned} &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs) \\ &= ((23.5 \cdot 0.5 - 10.2 \cdot 0.69) - (2.92) \cdot -0.25) / (0.69 - 0.5) \\ &= 29.0860 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

$$\begin{aligned} &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmax}) / (Xt - Xs) \\ &= ((23.5 \cdot 0.5 - 10.2 \cdot 0.69) - (2.92) \cdot 4.05) / (0.69 - 0.5) \\ &= -37.7095 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:



$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 37.7 \text{ kgf/cm}^2$$

Allowable Tube Stress $[\sigma_t, A]$:
 $= S_{ot} = 1203.2018 \text{ kgf/cm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety $[F_s]$:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.0018 + -0.064 \cdot 0.012)9.27^4); 1.25)$
 $= 1.2500$ (Should be ≤ 2)

Determine the factor $[r_t]$:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2})/4$
 $= (\sqrt{19^2 + (19 - 2 \cdot 2)^2})/4$
 $= 6.0519 \text{ mm}$

Determine the factor $[C_t]$:
 $= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 1973947 / 2135)}$
 $= 135.0872$

Determine the factor $[F_t]$:
 $= k \cdot L/r$
 $= 0.8 \cdot 760 / 6.05$
 $= 100.4650$

Buckling Allowable Stress $[S_{tb}]$:
 $= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (2135 / 1.25) \cdot (1 - 100 / (2 \cdot 135))$
 $= 1072.973 \text{ kgf/cm}^2$ (Never greater than S_{ot})

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load $[W_t]$:
 $= \sigma_{t, \max} \cdot \text{Tube Area}$
 $= 37.7 \cdot 1.07$
 $= 40.28 \text{ kgf}$

Tube Metal Area $[A_t]$:
 $= \pi(d_o - t)t$
 $= \pi(19 - 2)2$
 $= 1.068 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength $[F_t]$:
 $= \pi \cdot t(d_o - t)S$
 $= \pi \cdot 2(19 - 2)1203$
 $= 1285.188 \text{ kgf}$

Groove Weld Strength $[F_g]$:
 $= 0.85 \cdot \pi \cdot a_g(d_o + 0.67 \cdot a_g)S_w$ (but not $> F_t$)
 $= 0.85 \cdot \pi \cdot 2(19 + 0.67 \cdot 2)1203$
 $= 1285.1880 \text{ kgf}$

Weld Strength Factor $[f_w]$:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 1203 / 1203$
 $= 1.000$

Design Strength Ratio $[f_d]$:
 $= F_d / F_t$
 $= 40.3 / 1285$
 $= 0.031$

Fillet Weld Strength $[F_f]$:
 $= 0.55 \cdot \pi \cdot a_f(d_o + 0.67 \cdot a_f)S_w$ (but not $> F_t$)
 $= 0.55 \cdot \pi \cdot 2(19 + 0.67 \cdot 2)1203$
 $= 845.7292 \text{ kgf}$



Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 2 \cdot (19 - 2) \cdot 1 \cdot 0.031} - 0.75 \cdot 19 \right)$$

$$= 0.080 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 0.0400 mm

Minimum Required Groove Weld Leg [agr]: 0.0400 mm

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 1407 \cdot 1 = 1407 \text{ kgf/cm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 928^2 / ((1900 + 18) \cdot 18) \cdot [2.92 + (1.02^2 - 1)(23.5 - 10.2)] + 950^2 \cdot 10.2 / ((1900 + 18) \cdot 18)$$

$$= 354.9660 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 18 mm, the minimum Shell length adjacent to the tubesheet should be 332.878 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 928^2 / ((1900 + 18) \cdot 18) \cdot [2.92 + (1.02^2 - 1)(23.5 - 10.2)] + 950^2 \cdot 10.2 / ((1900 + 18) \cdot 18)$$

$$= 354.9660 \text{ kgf/cm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot ks / ts^2 \cdot \{ \beta_s [\delta_s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) \cdot (ao/h)^3 \cdot [1 + h \cdot \beta_s / 2] \cdot [Pe(Z_v + Z_m \cdot Q_1) + 2/ao^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 20627920 / 18^2 \cdot \{ 0.0098 [0.021626027 \cdot 23.5 + 950^2 \cdot 0 / (35531048)] + 6(1 - 0.41^2) / (402743) \cdot (928/60)^3 \cdot [1 + 60 \cdot 0.0098 / 2] \cdot [2.92(0.012 + 0.15 \cdot -0.032) + 2/928^2 \cdot 0.15 \cdot -399] \}$$

$$= 2016.2512 \text{ kgf/cm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$$

$$\therefore |355| + |2016| \leq 2111 \text{ kgf/cm}^2$$

$$\therefore 2371, \text{ must be } \leq 2111 \text{ kgf/cm}^2 \text{ (See Elastic-Plastic iteration below)}$$

Note:

For a given Channel thickness of 18 mm, the minimum Channel length adjacent to the tubesheet should be 332.878 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= ac^2 \cdot Pt / ((Dc + tc) \cdot tc \cdot \cos(\theta))$$

$$= 950^2 \cdot 10.2 / ((1900 + 18) \cdot 18 \cdot \cos(0))$$

$$= 266.573 \text{ kgf/cm}^2$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot \{ (\delta_c \cdot Cm \Delta + \beta_c - \zeta_c \cdot Cm \Theta) \cdot Pt - 6(1 - \nu^2) / (E \cdot h) \cdot (ao/h)^3 \cdot (Cm \Theta + h \cdot \beta_c / 2 \cdot Cm \Delta) \cdot [Pe(Z_v + Z_m \cdot Q_1) + 2/ao^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 21241204 / 18^2 \cdot \{ (0.021001635 \cdot 1 \cdot 0.0098 - 0.000 \cdot 1) \cdot 10.2 - 6(1 - 0.41^2) / 402743 \cdot (928/60)^3 \cdot [1 + 60 \cdot 0.0098 / 2 \cdot 1] \cdot [2.92(0.012 + 0.15 \cdot -0.032) + 2/928^2 \cdot 0.15 \cdot -399] \}$$

$$= 699.9621 \text{ kgf/cm}^2$$

Channel Stress Summation vs. Allowable

$$\therefore |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$$



:: | 267 | + | 700 | ≤ 2111 kgf/cm²
 :: 967 must be ≤ 2111 kgf/cm²

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	744.7	≤ 2110.7 kgf/cm ²	Pass
Tubesheet Shear Stress	94.0	≤ 1125.7 kgf/cm ²	Pass
Maximum Tube Stress	37.7	≤ 1203.2 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-37.7	≤ -1073.0 kgf/cm ²	Pass
Maximum Force on any one Tube	40.3	≤ 1285.2 kgf	Pass
Axial Membrane Stress in Shell	355.0	≤ 1407.1 kgf/cm ²	Pass
Shell Stress (jt. inter.)	2371.2	≤ 2110.7 kgf/cm ²	See below
Channel Stress (jt. inter.)	966.5	≤ 2110.7 kgf/cm ²	Pass

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(2319, 4638/2) = 2319 kgf/cm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(2478, 4956/2) = 2478 kgf/cm²

Compute Factor [facts]:
 = min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(2016) / 2319], 1) = 1

Compute Factor [factc]:
 = min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(700) / 2478], 1) = 1

Note: Since facts & factc are both ≥ 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	2371.2	≤ 4637.7 kgf/cm ²	Pass
Channel Stress (jt. inter.)	966.5	≤ 2110.7 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	33.291	60.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.040	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	0.040	2.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	43.099	...	1096	2111	196	1126 Ps+Pt-Th D1 Ok
D2uc	54.445	...	1747	2111	277	1126 Ps+Pt-Th D2 Ok
D3uc	33.291	...	745	2111	94	1126 Ps+Pt-Th D3 Ok
D4uc	13.827	...	79	2111	12	1126 Ps+Pt-Th D4 Ok



O1uc	16.908	...	764	4393	183	1126	Ps+Pt+Th	O1	Ok
O2uc	25.558	...	1323	4393	277	1126	Ps+Pt+Th	O2	Ok
O3uc	8.642	...	559	4393	94	1126	Ps+Pt+Th	O3	Ok
O4uc	19.000	...	...	4393	...	1126	Ps+Pt+Th	O4	Ok
D1c	40.269	...	895	2111	203	1126	Ps+Pt-Th-c	D1	Ok
D2c	51.674	...	1539	2111	290	1126	Ps+Pt-Th-c	D2	Ok
D3c	33.924	...	749	2111	100	1126	Ps+Pt-Th-c	D3	Ok
D4c	14.398	...	71	2111	13	1126	Ps+Pt-Th-c	D4	Ok
O1c	20.446	...	682	4393	190	1126	Ps+Pt+Th-c	O1	Ok
O2c	28.899	...	1302	4393	290	1126	Ps+Pt+Th-c	O2	Ok
O3c	14.060	...	620	4393	100	1126	Ps+Pt+Th-c	O3	Ok
O4c	19.000	...	...	4393	...	1126	Ps+Pt+Th-c	O4	Ok

Max: 54.4451 ... mm 0.828 0.258 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				:	Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd		
D1uc	102	1407	...	...		...	...	...	...		Ok
D2uc	242	1407	...	...		...	...	...	...		Ok
D3uc	355	1407	...	...		...	...	...	...		Ok
D4uc	11	1407	-11	-1017		...	...	...	...		Ok
O1uc	113	4637	...	...		...	...	...	...		Ok
O2uc	242	4637	...	...		...	...	...	...		Ok
O3uc	355	4637	...	...		...	...	...	...		Ok
O4uc	...	4637	...	...		...	...	...	...		Ok
D1c	117	1407	...	...		...	...	...	...		Ok
D2c	259	1407	...	...		...	...	...	...		Ok
D3c	388	1407	...	...		...	...	...	...		Ok
D4c	12	1407	-12	-1000		...	...	...	...		Ok
O1c	129	4637	...	...		...	...	...	...		Ok
O2c	259	4637	...	...		...	...	...	...		Ok
O3c	388	4637	...	...		...	...	...	...		Ok
O4c	...	4637	...	...		...	...	...	...		Ok
Max RATIO	0.276			0.012		...			...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	117	1203	-51	-1072	125	1285	838	2110	1761	2110	Ok
D2uc	148	1203	-148	-1072	158	1285	2876	4637	760	4955	Ok
D3uc	38	1203	-38	-1072	40	1285	2371	4637	967	2110	Ok



D4uc	7	1203	-3	-1072	7	1285	129	2110	34	2110	Ok
O1uc	110	2406	-47	-1072	118	2570	730	4637	1727	4955	Ok
O2uc	148	2406	-148	-1072	158	2570	2876	4637	760	4955	Ok
O3uc	38	2406	-38	-1072	40	2570	2371	4637	967	4955	Ok
O4uc	...	2406	...	...	...	2570	...	4637	...	4955	Ok
D1c	129	1203	-52	-1072	138	1285	1026	2110	2124	4955	Ok
D2c	161	1203	-161	-1072	172	1285	3219	4637	880	4955	Ok
D3c	39	1203	-39	-1072	42	1285	2572	4637	1204	2110	Ok
D4c	7	1203	-3	-1072	8	1285	145	2110	40	2110	Ok
O1c	122	2406	-48	-1072	130	2570	905	4637	2084	4955	Ok
O2c	161	2406	-161	-1072	172	2570	3219	4637	880	4955	Ok
O3c	39	2406	-39	-1072	42	2570	2572	4637	1204	4955	Ok
O4c	...	2406	...	...	...	2570	...	4637	...	4955	Ok

Max RATIO 0.134 0.150 0.134 0.694 0.834

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	54.445	60.000	Ok
Tube thickness :	0.479	2.000	Ok
Tube-tubesheet fillet weld leg :	0.170	2.000	Ok
Tube-tubesheet groove weld leg :	0.170	2.000	Ok

Min Shell length of thk, (18.000) adj. to tubesheet: 332.878 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 332.878 mm

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -.0125 mm
 Due to Pressure + Thermal Shell Compresses by : -.0125 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	21.576	0.996	28.791	0.999
Tubesheet Shear Stress	60.402	1.000	91.027	1.000
Tube Tensile Stress	100.841	1.000	175.419	1.000
Tube Compressive Stress	58.645	0.254	156.432	1.000
Tube-Tubesheet Joint load	100.841	1.000	175.418	1.000
Shell Stress (Axial, Junction)	21.576	0.996	28.581	0.993
Tubesheet-Channel Junction	21.576	0.996	28.581	0.993
Tube Pressure Stress	276.597	1.000	176.371	1.000
Minimum MAWP	21.576		28.581	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	21.576	0.999	28.581	0.999
Tubesheet Shear Stress	62.713	1.000	95.311	1.000
Tube Tensile Stress	111.089	1.000	190.446	1.000
Tube Compressive Stress	257.780	1.000	189.327	1.000
Tube-Tubesheet Joint load	111.089	1.000	190.446	1.000
Shell Stress (Axial, Junction)	21.576	0.999	28.581	0.999
Tubesheet-Channel Junction	21.576	0.999	28.581	0.999
Tube Pressure Stress	276.597	1.000	181.218	1.000
Minimum MAPnc	21.576		28.581	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)



Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0225	0.0225	...	...	...	...
2	-0.1253	-0.1254	...	...	...	...
3	-0.1085	-0.1086	...	...	...	...
4	0.0056	0.0056	...	...	...	...
5	0.0168	0.0168	...	...	...	...
6	-0.1253	-0.1254	...	...	...	...
7	-0.1085	-0.1086	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(60/4, \min(60, 18))$$

$$= 18.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(23.5/28.6, 1)$$

$$= 0.821$$

Thickness Ratio = 0.82, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(60/4, \min(60, 18))$$

$$= 18.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(10.2/21.6, 1)$$

$$= 0.473$$

Thickness Ratio = 0.47, Temperature Reduction per Fig. UCS 66.1 = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-47 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

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